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LUMINA

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Bachelor of Science in Computer Science (2022-2026)

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Executive Summary

Lumina is an AI-powered career counselling and academic guidance platform designed to modernize how students navigate their educational and professional journeys. Built as a mobile application, Lumina connects students, alumni, universities, and administrators in one unified ecosystem, ensuring that every learner receives the support, clarity, and real-time information needed to make confident decisions about their future.

At its core, Lumina addresses a key challenge: traditional counselling systems are often fragmented, outdated, and inaccessible to many students. Lumina transforms this experience by integrating artificial intelligence, data-driven insights, and user-friendly interfaces to deliver personalized guidance at scale. Students can explore universities, predict their admission chances through intelligent merit algorithms, receive career recommendations, interact with an AI chatbot for instant answers, and stay updated with the latest admission news all within a single application.

The platform also encourages meaningful engagement through community features. Students can interact with alumni for mentorship, seek help from peers, share insights, and build networks that extend beyond academic boundaries. Universities, on the other hand, can use Lumina to communicate announcements, provide transparent admissions details, and gain valuable analytics about student interests and trends.

From a technical perspective, Lumina is developed with modern, scalable technologies. Flutter ensures smooth cross-platform performance, Firebase powers secure and real-time backend operations, and advanced machine learning models enable intelligent recommendations and prediction systems. This combination allows Lumina to remain responsive, accurate, and adaptable as the user base grows.

Ultimately, Lumina aims to redefine the academic and career planning experience by offering a smart, connected, and accessible digital counselling ecosystem. By bridging the gap between students, alumni, and institutions, it empowers learners to make informed choices, enhances transparency within the education sector, and supports long-term career readiness in a rapidly evolving world.

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Arham Khaliq	Nihal Tariq	Ramisha Rahat
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Abbreviations

Abbreviations	**Full Form**
KPI	Key Performance Index
RBAC	Role-Based Access Control
SRS	Software Require Specification
PC	Personal Computer
AI	Artificial Intelligence
FYP	Final Year Project
UI/UX	User Interface/User Experience
UAT	User Acceptance Testing
UCD	User Centered Design
API	Application program Interface
HCI	Human Computer Interaction
ML	Machine Learning
DSA	Data Structure & Algorithm
NLP	Natural Language Process
SDLC	Software development life cycle
UAT	User acceptance Testing
SDD	Software Design & Development
FR/NFR	Functional Requirements/Non-Functional Requirements

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Chapter 1

INTRODUCTION

1. Introduction

In today’s world, where academic pathways and career opportunities are expanding rapidly, students face more pressure than ever to make the right decision one that will shape their long-term careers. Choosing the ideal university, selecting a suitable degree program, understanding admission requirements, and meeting strict deadlines are all complex tasks that can easily overwhelm young individuals. Every student hope to find an institution that aligns not only with their academic strengths but also with their financial limitations and future professional goals. However, the abundance of information available, combined with its scattered nature, often makes the process confusing rather than helpful.

The rising complexity of the modern educational landscape has turned the university admission process into a significant source of stress. Students must navigate countless institutional websites, sift through varying merit cutoffs from year to year, and compare hundreds of degree programs with subtle differences in curriculum and outcome. This fragmented, information-heavy environment often penalizes students from less privileged backgrounds or those without immediate access to experienced academic counselors, leading to decisions based on guesswork, peer pressure, or incomplete data. This systemic challenge necessitates a powerful, centralized solution that democratizes access to quality, personalized career advice [1].

Recognizing these challenges, we developed Lumina, an innovative AI-powered career guidance and academic support platform that simplifies this entire decision-making journey. Lumina is designed to provide students with personalized university recommendations, merit-based predictions, and strategic academic insights, enabling them to make confident and well-informed decisions about their future. Beyond academic guidance, Lumina fosters a supportive community where students can interact with peers and alumni, share experiences, and seek mentorship. The platform also integrates an AI-driven chatbot that not only assists with queries but also contributes to student well-being by offering empathetic support and reducing the stress associated with academic planning.

In an educational environment that evolves rapidly, students often struggle to stay updated with admission policies, shifting merit cutoffs, and institution-specific criteria. The lack of centralized and reliable information makes the process even more challenging. Many students rely on outdated resources, hearsay, or guesswork when selecting universities, which can lead to misguided decisions or missed opportunities. Additionally, predicting whether one’s scores meet the requirements of a specific institution is difficult due to inconsistent data available across various platforms.

Lumina addresses these gaps by functioning as a comprehensive, AI-driven career counselling system. It integrates machine learning models, real-time data updates, and predictive analytics to offer customized university suggestions tailored to each student's academic profile, interests, and financial considerations. The platform also maintains a continuously updated database of merit trends, deadlines, and admission requirements, ensuring that students always have access to accurate, reliable, and up-to-date information.

By combining personalized recommendations, real-time data, and community-driven support, Lumina empowers students to approach their academic future with clarity and confidence. It transforms a traditionally overwhelming process into a structured, intelligent, and user-friendly experience ultimately guiding students toward better decisions and brighter career outcomes.

1.1 Purpose

Lumina is a mobile application designed to assist students in making informed decisions about university admissions and career choices. The primary goal of Lumina is to provide students with a centralized platform where they can:

- Get personalized university recommendations based on academic scores, financial situation, and interests.
- Use an AI-driven merit prediction model to estimate their chances of admission.
- Access real-time university admission updates and deadlines.
- Interact with a community of students to share experiences and reviews.
- Seek career guidance through an AI-powered chatbot.

At the same time, universities can:

- Advertise their programs, courses, and scholarships.
- Engage with students through post updates about admissions.

Lumina aims to become a one-stop career counseling solution that simplifies decision-making for students while offering universities a direct communication channel to reach prospective applicants.

1.2 Brief

Lumina is an AI-powered career guidance and university counselling application designed to help students make informed and confident academic decisions with ease. The project addresses major challenges students typically face during the admission process such as selecting the right university, understanding merit requirements, comparing programs, and staying updated with critical deadlines. By integrating advanced tools like Flutter [4] for frontend development, Firebase [5] for backend management, and Google Gemini AI for intelligent decision-making, Lumina delivers personalized university recommendations, accurate merit predictions, and program comparisons tailored to each student's academic profile.

The core innovation of Lumina lies in its intelligent data processing capabilities. Traditional guidance relies on static brochures; Lumina uses machine learning to analyze historical merit data, program popularity, and student performance metrics to generate predictions that are far more accurate and dynamic. This predictive insight minimizes the anxiety of the application process by giving students a realistic estimate of their admission probability. Furthermore, the application is built with a Clean Architecture approach to ensure the system is maintainable, testable, and robust against future changes in technology or data requirements.

Beyond academic guidance, Lumina also provides real-time insights and a dedicated community space where students, alumni, and mentors can connect, share experiences, and offer support. The inclusion of an AI-based therapy chatbot further strengthens the platform by promoting student mental well-being a key factor often overlooked in traditional counselling systems. The outcome is a fully functional prototype that combines intelligent algorithms, clean UI/UX design, and emotional support tools. This report covers the system's overall architecture, design strategies, implementation details, methodological approaches, and evaluation results, giving a complete overview of the decisions and technologies used throughout development.

The Lumina application is structured into a set of distinct, interconnected modules, each handling a specific set of core functionalities to cater to the diverse needs of its user base: Students, Alumni, and Universities, as well as the administrative roles.

1.2.1 Authentication Module

This module is the security gateway for the entire application, responsible for securely identifying and verifying all users before granting access.

- **Registration/Sign-Up:** Allows new users (Students, Alumni, University Reps) to create an account using their email and a secure password.
- **Login/Sign-In:** Validates the credentials of existing users to grant them access.
- **Email Verification:** Implements a mechanism to confirm the validity of a user's email address upon registration, enhancing security and data cleanliness.
- **Password Recovery:** Provides a secure way (e.g., via email link) for users to reset forgotten passwords.

1.2.2 User Categorization Module

This module is crucial for organizing the user base and dictating the level of access and available features for each user.

- **Role Assignment:** Automatically or manually assigns a specific role upon registration: Student, Alumnus, or University Representative.
- **Role-Based Access Control (RBAC):** Ensures that users only see and interact with features relevant to their assigned role (e.g., a student cannot post an official admission notice, only a University Representative can).
- **Privilege Management:** Defines and manages the specific permissions associated with each user category across all other modules.

1.2.3 Profile Module:

This module handles the creation, storage, and management of user data, which is essential for personalization and community interaction.

- **Creation and Update:** Allows users to input and edit their profile information.
  - **Students:** Name, academic scores, educational history, financial preference, career interests.
  - **Alumni:** University attended, graduation year, work experience, area of expertise.
  - **Universities:** Official name, location, contact details, accredited programs, institution type.
- **Viewing:** Enables other users to view a public version of the profile (e.g., Students viewing an Alumnus's experience).
- **Privacy Settings:** Gives users control over which parts of their profile are visible to the community.

1.2.4 User Post Module:

This module facilitates the community interaction features of Lumina, allowing users to share, discuss, and engage with content.

- **Content Creation:** Enables Students and Alumni to post questions, share experiences, articles, and career tips. Universities can post official admission updates, deadlines, and scholarship notices.
- **Interaction Features:** Allows users to like, comment on, and share posts.
- **Following/Bookmarks:** Students can follow specific universities or Alumni for updates, and **bookmark** posts for future reference.
- **Real-Time Feed:** Displays a personalized stream of content based on the user's follows, interests, and profile data.

1.2.5 Chatbot AI Module:

The core AI component that provides instant support, guidance, and well-being features.

- **Query Resolution:** Answering common questions about admission processes, university requirements, and degree programs using the knowledge base.
- **Empathetic Support:** Utilizes AI to act as a supportive, first-line mental wellness tool, offering empathetic responses and stress reduction advice related to academic pressure.
- **Information Retrieval:** Acts as a smart search tool, quickly pulling up facts and data from the Lumina database.
- **Conversation Logging:** Securely logs interactions for quality improvement and to train the model over time.

1.2.6 User Communication Module (One-on-One and Groups):

This module supports direct, secure communication channels within the platform.

- **Direct Messaging (One-on-One):** Enables real-time chat between Students and Alumni for mentorship, or between Students and University Representatives for official inquiries.
- **Group Chat/Community Discussions:** Allows Students to form topic-based groups or for the system to host moderated community forums for broader discussions.
- **Notification Integration:** Provides instant notifications for new messages to ensure timely responses.

1.2.7 University Recommendation Module:

This is a key differentiator of Lumina, providing personalized and data-driven guidance.

- **Personalized Suggestions:** Uses a machine learning model to analyze the student's profile (scores, interests, financial ability) against the University and Program database.
- **Merit Prediction Model:** Based on historical data, the AI predicts the probability of a student meeting the cutoff for a specific program, giving a clear, actionable estimate of their chances.
- **Filtering and Comparison:** Allows Students to filter universities by criteria (location, program, fee structure) and directly compare programs side-by-side.
- **Real-Time Updates:** Integrates the latest admission and merit data to ensure recommendations are always current.

1.2.8 Admin Module (University Management):

This module is designed for system administrators to maintain the application's integrity, data quality, and security.

- **User Management:** Ability to view, verify, suspend, or delete user accounts (especially for University and Alumni verification).
- **Content Moderation:** Tools to review and manage user-generated content (posts, comments) to maintain a safe and respectful community environment.
- **System Monitoring:** Dashboard to track key performance indicators (KPIs) like active users, system load, and error logs.
- **Data Updates:** Central tool for importing, validating, and updating the core database of universities, programs, and historical merit data.
- **Analytics:** Provides reports to the Lumina team and non-sensitive aggregated data to university partners (e.g., student interest metrics).

1.3 Relevance to Course Modules

The development of Lumina is strongly rooted in the academic foundation provided throughout the Bachelor of Computer Science (BCS) program. Concepts learned in Mobile Application Development played a direct role in creating the app's frontend using Flutter, while principles from Software Engineering guided the project's planning, requirement analysis, design methodology, and testing. Knowledge from Database Systems and Cloud Computing was applied in designing Firebase-based backend structures, ensuring secure authentication, efficient data flow, and real-time synchronization for notifications, messages, and admission updates.

The project also draws heavily from Artificial Intelligence, where machine learning, recommendation systems, and NLP concepts were used to build personalized recommendation algorithms, merit prediction logic, and the AI chatbot. Furthermore, Human–Computer Interaction (HCI) contributed significantly to designing an intuitive UI/UX that prioritizes ease of use and accessibility for students. Supporting modules like Web Technologies, Programming Fundamentals, and Data Structures & Algorithms helped ensure optimized code practices, smooth performance, and correct implementation of backend logic. Overall, Lumina serves as a practical demonstration of how multiple theoretical modules from the BCS curriculum combine to create a complete, real-world AI-driven mobile application.

1.3.1 Mobile Application Development:

This module was crucial as it provided the technical skills necessary to build the application's user-facing elements. Knowledge of mobile frameworks, specifically Flutter, was applied to develop the cross-platform frontend. This allowed Lumina to offer a consistent, high-quality user experience across both Android and iOS devices. The module's concepts were instrumental in managing the State of the application, handling user gestures, and ensuring the UI dynamically responds to real-time data updates from the server, which is vital for the chat and notification features.

1.3.2 Software Engineering:

Principles from Software Engineering provided the methodological foundation for the entire Final Year Project (FYP). This guided the disciplined approach used for requirements analysis (defining functional and non-functional needs), system design methodology (choosing an appropriate architecture like Clean Architecture [12]), and project planning (scheduling tasks and milestones). Most importantly, it dictated the implementation of rigorous testing procedures—such as unit testing and integration testing—to ensure the application is reliable, maintainable, and scalable to support a growing student user base and increasing university data.

1.3.3 Database Systems and Cloud Computing:

This module was essential for creating the application's secure and scalable backend infrastructure. Concepts from Database Systems were applied to design the structured data models (schemas) for users, universities, programs, and posts within Firebase Firestore. Knowledge of Cloud Computing, specifically using Firebase, was vital for implementing key backend services: secure user authentication, efficient data retrieval and storage, and the use of services like Cloud Functions for executing backend logic (e.g., triggering notifications). This combination ensures real-time synchronization for chat messages, admission updates, and notifications.

1.3.4 Artificial Intelligence (AI) and Machine Learning (ML):

AI and ML concepts form the **intelligent core** of the Lumina platform. Principles were directly applied to three key areas:

- **Recommendation Systems:** Machine learning techniques (like collaborative or content-based filtering) were used to analyze student profile data (scores, interests) against historical trends to provide highly personalized university recommendations.
- **Merit Prediction Logic:** Supervised learning models were trained on historical merit cutoffs to create the predictive algorithms that estimate a student's chances of admission, turning static data into actionable insight.
- **NLP and Chatbot:** Natural Language Processing (NLP) skills were used to interpret user queries and power the AI-driven chatbot (using Google Gemini), allowing it to provide relevant answers and empathetic support.

1.3.5 Human-Computer Interaction (HCI):

HCI provided the theoretical basis for designing the User Interface (UI) and User Experience (UX). The focus was on usability and accessibility. Concepts like user profiling, information architecture, and intuitive navigation were applied to organize complex academic and career information in a simple, non-overwhelming manner. By prioritizing clear visual hierarchy and reducing cognitive load, the platform makes the difficult process of academic decision-making accessible to all students.

1.3.6 Data Structures and Algorithms (DSA):

Efficient data handling and system speed are dependent on DSA knowledge. This was applied to optimize several performance-critical functions:

- **Searching and Filtering:** Algorithms (like binary search or efficient hashing techniques) were used to enable quick searching and filtering of university listings, programs, and community posts, even with a large dataset.

- **Data Structures:** Appropriate structures (e.g., hash maps for quick user lookups, linked lists for managing notification queues) were chosen to handle user profiles, community feeds, and recommendation results efficiently. This directly improves app performance and overall user experience.

1.3.7 Programming Fundamentals and Web Technologies:

These foundational modules ensured the successful technical implementation of the project. Programming Fundamentals guaranteed the use of clear, modular, and maintainable code across the application. Web Technologies knowledge was critical for handling the client-server communication, specifically dealing with API requests and data formatting (like JSON), which facilitates the smooth exchange of information between the Flutter frontend and the Firebase backend services.

1.4 Project Background

Lumina is envisioned as a transformative solution that leverages Artificial Intelligence to simplify and modernize the complex process of academic and career decision-making for students. Traditionally, students have relied on fragmented sources manual research, hearsay, limited counselling services, or outdated websites to identify suitable universities or choose career pathways. These conventional methods are not only time-consuming but also lack personalization, leaving students uncertain about whether their choices align with their abilities, aspirations, and financial constraints.

Lumina addresses these issues by introducing an AI-driven ecosystem that analyzes student profiles, academic performance, merit trends, and institutional requirements to generate data-backed university recommendations. The platform integrates Natural Language Processing (NLP) technologies to power an intelligent chatbot that offers real-time guidance, academic suggestions, and mental health support. Students can interact with the chatbot for academic inquiries, degree advice, or emotional reassurance.

Through the combination of mobile application development, cloud-based databases, and machine learning models, Lumina represents a practical demonstration of how intelligent systems can revolutionize the counselling process. It transforms a traditionally overwhelming experience into one that is structured, accessible, and deeply personalized, making academic guidance more efficient and inclusive for a diverse range of students.

1.5 Literature Review

The evolution of career guidance platforms has seen a shift from static information portals to interactive, data-driven systems capable of providing tailored insights. In Pakistan, commonly used platforms such as EduVision [2] and IlmKiDuniya [3] offer extensive lists of universities, degree programs, and admission criteria. While informative, these platforms typically suffer from several drawbacks, including outdated user interfaces, cluttered advertisements, and navigation challenges that hinder quick access to critical information. Additionally, their web-only nature limits convenience, particularly for students who rely heavily on mobile devices.

Globally, educational technology has moved towards AI-enhanced platforms designed to provide personalized learning pathways, predictive analytics, and mental health support. Many international systems use machine learning algorithms to match students with universities, assess their skills, and forecast academic success. However, these platforms frequently cater to international education systems and lack localized features such as Pakistani merit calculations, quota systems, or real-time admission tracking.

Students also turn to social media platforms, online forums, and peer-review sites for guidance. While these sources offer diverse opinions, they lack structure, authenticity, and reliability. Information is often scattered, unverified, or contradictory, which further complicates academic decision-making for students.

Given the growing use of smartphones among youth, there is a strong demand for a mobile-first solution that integrates verified academic information, AI-driven prediction models, community interaction, and emotional support in one cohesive environment. Lumina positions itself to fill this gap by offering personalized, intelligent, and mobile-centric guidance tailored to Pakistani students.

1.6 Analysis from Literature Review

The literature review **Table 1.1: Analysis of literature review** highlights several gaps in current educational guidance solutions that shaped the vision and design of Lumina. Traditional platforms provide basic academic information but fail to incorporate personalization, mobile accessibility, predictive analytics, or emotional well-being support. Their dependency on web browsers and heavy reliance on advertisements negatively impacts user experience and accessibility.

These identified shortcomings directly influenced Lumina's development approach. Lumina introduces a modern, minimalistic, and ad-free interface that prioritizes ease of navigation and fast access to information key features lacking in existing systems. By integrating machine learning algorithms for merit prediction and university recommendation, Lumina differentiates itself from typical information portals.

Moreover, the inclusion of an AI-powered therapy and guidance chatbot addresses a crucial yet neglected area in existing solutions: student mental health. Global trends increasingly emphasize holistic student support, and Lumina adopts this approach while localizing it to the needs of Pakistani students.

In essence, the literature emphasizes the absence of an integrated, AI-enabled, mobile-first platform. Lumina fills this void by combining intelligence, personalization, real-time data, and emotional support, providing a comprehensive, modernized alternative to existing platforms.

Globally, educational technology has moved toward AI-enhanced platforms designed to provide personalized learning pathways, predictive analytics, and even mental health support. Many international systems use machine learning algorithms to match students with universities, assess their skills, and forecast academic success. However, these platforms frequently cater to international education systems and lack localized features critical for Pakistani students, such as specific merit calculations, understanding institutional quota systems, or tracking real-time local admission deadlines.

Students also frequently turn to social media platforms, online forums, and peer-review sites for guidance. While these sources offer diverse opinions and a sense of community, they inherently lack structure, authenticity, and reliability. Information is often scattered, unverified, or contradictory, which further complicates academic decision-making.

Given the pervasive use of smartphones among youth, there is a strong demand for a mobile-first solution that integrates verified academic information, AI-driven prediction models, robust community interaction, and emotional support in one cohesive environment. Lumina positions itself to fill this market gap by offering personalized, intelligent, and mobile-centric guidance tailored specifically to the needs of Pakistani students. The following table summarizes the weaknesses of existing platforms in the context of Lumina's proposed features.

Table 1.1: Analysis of literature review

Application/Website Name	**Weakness**	**Proposed Project Solution (Lumina)**
EduVision (Web)	Static, general information; outdated	Offers a modern,

	UI/UX; heavy reliance on advertisements; limited mobile optimization.	minimalistic, ad-free mobile-first UI/UX with real-time, dynamic data.
IlmKiDuniya (Web)	Lack of personalized recommendation; general admission criteria only; no predictive analytics; slow updates.	Integrates AI-driven merit prediction and personalized university recommendations based on student profile.
Career Guidance PK (Mobile App)	Focus on basic course information; lacks community forum; does not integrate AI for personalized advice or emotional support.	Includes a dedicated Alumni/Peer Community Module and an AI Chatbot for guidance and emotional support.
ICS Career GPS: Guide for all (Mobile App)	Primarily skill/interest assessment; limited focus on localized, real-time university admission and merit tracking.	Provides highly localized data, addressing Pakistani merit calculations and real-time university admission deadlines.

The literature review robustly highlights several key gaps in current educational guidance solutions that were instrumental in shaping the vision and detailed design of Lumina. Traditional platforms, both local and international, typically provide a necessary but ultimately incomplete set of services. They offer basic academic information but fundamentally fail to incorporate the critical elements of personalization, mobile accessibility, predictive analytics, and holistic emotional well-being support. Their dependency on web browsers and heavy reliance on distracting advertisements negatively impacts user experience and accessibility, particularly for the target youth demographic.

The identified shortcomings served as the direct mandate for Lumina's innovative development approach. Lumina is designed to be a definitive solution, starting with a modern, minimalistic, and ad-free mobile interface that prioritizes ease of navigation and fast access to critical information a key feature conspicuously lacking in existing systems. By integrating advanced machine learning algorithms for merit prediction and personalized university recommendation, Lumina moves beyond being a simple information portal to become a true intelligent counselling system. Moreover, the inclusion of an AI-powered guidance and wellness chatbot addresses a crucial yet neglected area: student mental health. Global EdTech trends increasingly emphasize holistic student support, and Lumina adopts this approach while localizing it specifically to the socio-academic needs of Pakistani students. In essence, the literature review confirms the critical market need for an integrated, AI-enabled, mobile-first platform, and Lumina is precisely engineered to fill this void by combining intelligence, personalization, real-time data, and emotional support.

1.6.1 Addressing Lack of Personalization

Existing platforms predominantly operate on a one-size-fits-all model, where students are presented with a massive, unfiltered list of universities and programs. This forces the student to manually cross-reference their scores, interests, and financial capacity against the provided static data. Lumina overcomes this by integrating a sophisticated Recommendation of University Module. This system uses collaborative filtering and predictive analytics to generate a ranked list of suggestions that are highly specific to the individual student's academic profile, thereby drastically reducing the time and mental effort required for decision-making.

1.6.2 Improved Categorization and UI/UX Design

The documented weakness of outdated UI/UX and cluttered advertisements in local web portals directly informed Lumina's design strategy. Following principles from HCI, Lumina employs a clean, minimalistic design that is optimized for the mobile screen. The focus is on clear categorization and an intuitive information hierarchy, ensuring that students can reach vital data (deadlines, program details, merit trends) within a maximum of three taps. This design choice guarantees a fast, frustration-free experience, prioritizing the student's focus on the crucial admission information.

1.6.3 Integration of Predictive Analytics for Merit

A significant functional gap in current platforms is the absence of predictive analytics. Students are left to guess their chances of admission based on the previous year's closing merit, which is often unreliable. Lumina directly solves this through its AI-driven Merit Prediction Model. This module utilizes machine learning to analyze years of historical merit data, institutional admission criteria, and current application trends to provide a calculated, probability-based estimate of a student's admission chances. This level of data-backed forecasting drastically increases the confidence and effectiveness of the student's application strategy.

1.6.4 Integration of Chat Support and Real-Time Interaction

The need for real-time support and a community ecosystem is evident from the reliance on unreliable social media forums. Lumina's solution is two-fold:

- AI Chatbot Module: Provides instant, structured guidance and emotional support, acting as the first line of empathetic counselling.
- User Communication Module (Texting/Groups): Fosters a genuine, verifiable community by connecting students with credible Alumni/Mentors and peers, ensuring that shared experiences are both supportive and reliable, thus filling the gap of unstructured, unverified social media advice.

1.6.5 Mobile-First and Localized Focus

The literature highlighted that most comprehensive systems are not localized for the Pakistani education system or lack a robust mobile app. Lumina is inherently a mobile application, built with Flutter, ensuring universal accessibility for the youth who primarily use smartphones. Crucially, the data engine is specifically configured to manage and track Pakistani merit calculations, institution-specific quota systems, and real-time local admission cycles, making it the most relevant and efficient guidance tool for the target demographic.

1.7 Methodology and Software Lifecycle for This Project

The development of Lumina followed a structured and modern software engineering approach that combined the Agile [7] Software Development Life Cycle (SDLC) with the Iterative Model. Since Lumina is a dynamic, feature-rich, and scalable application involving AI integration, real-time chat, predictive models, and multiple user roles, a flexible methodology was essential. Agile was chosen due to its adaptability, frequent feedback cycles, and ability to handle evolving user needs. The Iterative Model was used alongside Agile to ensure that Lumina was built in phases, with each version becoming more refined, functional, and optimized.

The Agile approach enabled the project to be divided into multiple short, manageable sprints, where each sprint focused on developing a specific feature set such as user registration, community posting, alumni chat, or AI-based recommendation modules. After each sprint, the implemented functionalities were tested, reviewed, and improved based on user feedback and performance evaluation. This ensured that the final system was not only functional but also user-centered, stable, and aligned with real-world requirements.

Alongside Agile, the Iterative Model helped break the system into multiple cycles, starting with essential features and gradually expanding into advanced functionalities. In each iteration, previously built components were revisited, enhanced, and optimized. This approach was especially effective for AI-related modules such as merit prediction, recommendation engines, and chatbot interaction as these components required continuous tuning and data validation. Through combined Agile Iterative development, Lumina evolved through continuous refinement, ensuring consistency, scalability, and high-quality performance across all modules.

1.7.1 Rationale behind Selected Methodology

1.7.1.1 Suitability of Agile Methodology

Agile was selected because Lumina required regular updates, continuous feedback, and flexible adjustments throughout its development cycle. The system involves diverse, highly dependent modules real-time chat, university data integration, role-based access, push notifications, and AI algorithms each of which demanded repeated testing and improvement.

- **Allows frequent changes:** Requirements for academic data and user expectations are dynamic. Agile allowed the team to incorporate changes based on testing or stakeholder feedback without derailing the entire project schedule.
- **Supports incremental development:** This ensured that core modules (like user authentication and profile creation) were delivered early and then progressively enhanced with more advanced features.
- **Encourages parallel development:** UI/UX design, backend logic, AI model training, and database configurations could progress simultaneously, speeding up overall development time.
- **Improves communication:** Daily meetings (stand-ups), reviews, retrospectives, and sprint planning ensured constant communication and transparency within the development cycle.
- **Reduces development risk:** Issues were identified and fixed early within each short sprint, preventing minor bugs from becoming major system failures later on.

Agile ultimately ensured that Lumina remained flexible, user-focused, and adaptable to changing academic data, user expectations, and technological requirements.

1.7.1.2 Suitability of the Iterative Model

The Iterative Model complemented Agile by ensuring that Lumina was built in phases, where every version became functionally better than the last. Instead of waiting until the final stage to test the entire system, the Iterative approach allowed for early prototyping and repeated refinement of specific components.

- **Allows early prototyping:** The project could start with basic, functional prototypes, enabling early user testing and feedback on the core concept before investing heavily in complex features.
- **Reduces risk through testing:** Enabled early detection and resolution of fundamental design or logic issues within specific modules (e.g., initial database structure or core chat functions).
- **Perfect for AI-driven systems:** AI models (merit prediction, recommendation engine) require constant data validation, testing, and repeated refinement; the Iterative cycle provided the structure for this continuous enhancement.
- **Enables smooth integration:** Allowed for the progressive integration of complex features like the AI therapy chatbot, predictive algorithms, and alumni matching, ensuring they seamlessly connected with the existing core system.

Using Agile and Iterative together allowed Lumina to evolve naturally starting with core authentication and database setup, progressing to recommendations and real-time features, and finishing with optimization, security, and final analytics.

1.7.2 Implementation of Agile + Iterative Lifecycle in Lumina

The Lumina project was executed in four distinct iterations, each building upon the stability and functionality established in the previous phase:

- **First Iteration: Foundation Development**

a) **Focus:** Core security and user access.

b) **Features:** User registration and authentication using Firebase Auth; initial implementation of RABC for students, alumni, and universities; creation of basic UI screens and navigation; and the establishment of the initial database structure and cloud storage setup.

- **Second Iteration: Core Functionalities**

a) **Focus:** Basic platform interaction and community.

b) **Features:** Implementation of the University listing and filtering feature; development of the Community forum for posting and replying; integration of the Alumni networking and basic chat functionality; and completing the backend integration with university datasets.

- **Third Iteration: AI Features and Enhancements**

a) **Focus:** Introducing intelligence and real-time data.

b) **Features:** Development and training of the AI-powered university recommendation engine; integration of the AI therapy chatbot using Google Gemini; and implementation of the real-time system for delivering admission updates from followed universities.

- **Fourth Iteration: Optimization & Advanced Features**

a) **Focus:** Polishing, performance, and final delivery.

b) **Features:** Implementation of Push Notifications with FCM; development of bookmarking and personalized alerts; comprehensive performance optimization and data security enhancements; final UI/UX refinement based on user testing; and execution of final system testing, debugging, and documentation.

1.7.3 Methodology Summary:

The Lumina project adopted a hybrid Agile-Iterative development methodology to ensure flexibility, high quality, and continuous alignment with user needs. The Agile SDLC provided the framework for organizing work into short, manageable sprints, facilitating early and frequent delivery of functional modules. This allowed the development team to quickly incorporate user and stakeholder feedback, making the project highly adaptable to evolving academic data and functional requirements. Concurrently, the Iterative Model ensured the system was built in incremental phases, starting with core authentication and gradually expanding to complex features. This approach was particularly beneficial for developing the AI-driven components (merit prediction and recommendation engines), which required repeated refinement and continuous validation with data. The combined methodology mitigated development risks, enhanced communication, and ultimately delivered a robust, scalable, and user-centered platform through a structured, four-iteration development cycle.

Chapter 2

PROBLEM DEFINITION

2. Problem Definition

In a constantly evolving educational landscape, students face tremendous difficulty navigating academic choices, selecting universities, predicting admission chances, and keeping track of critical deadlines. The absence of centralized, reliable, and easy-to-understand information often forces them to depend on outdated websites, unverified social media posts, or informal counselling that lacks accuracy.

Existing platforms in Pakistan, although informative, suffer from usability issues such as overwhelming advertisements, outdated designs, lack of personalization, and weak mobile accessibility. Meanwhile, no existing system offers integrated emotional or mental support even though academic pressure is a major source of stress for students.

Thus, the core problem is the lack of a mobile-first, AI-powered, unified solution that offers personalized academic guidance, reliable recommendations, real-time updates, and mental health support in one accessible platform. Lumina was developed specifically to overcome these challenges.

2.1 Problem Statement

Students today face significant challenges in making informed academic and career decisions due to the lack of centralized, reliable, and personalized guidance. The existing methods such as browsing multiple web portals, relying on outdated counselling centers, or seeking advice from peers are often fragmented and time-consuming. Platforms like EduVision and IlmKiDuniya provide useful information, but they are primarily web-based, cluttered with advertisements, and lack intelligent features that can adapt to the individual needs of students. As a result, students struggle to identify universities and degree programs that align with their academic strengths, interests, financial capabilities, and long-term career aspirations.

In addition to the absence of personalized recommendations, students also face difficulties in predicting their chances of admission. Merit cutoffs, admission policies, and university criteria vary annually, making it challenging to estimate eligibility accurately. The scattered nature of this information forces students to search across multiple unreliable sources, increasing confusion and reducing the likelihood of making timely decisions. Moreover, amid academic stress and uncertainty, students often lack emotional support or guidance tools, which further impacts their decision-making process.

Therefore, the core problem lies in the absence of a mobile-first, AI-powered platform capable of integrating academic recommendations, merit predictions, real-time updates, and emotional support in one unified system. Students need a smart and accessible solution that not only provides accurate information but also understands their unique needs and guides them throughout their academic journey. Lumina aims to address these gaps by offering intelligent counselling, predictive analytics, community interaction, and mental well-being support—all within a user-friendly mobile application.

2.2 Deliverables and Development Requirements

The successful completion of the Lumina project is defined by a set of clear, functional outputs (Deliverables) and the strict adherence to modern technology standards and protocols (Development Requirements). These elements collectively ensure that Lumina effectively addresses the identified need for a robust, scalable, and AI-driven academic guidance platform. The final delivery of the Lumina project encompasses both the functional software components and the supporting documentation necessary for its deployment, maintenance, and future development.

2.2.1 Core Functional Applications

The primary deliverable is a fully functional mobile application developed using Flutter, integrating all defined AI and community features. This includes:

- A Personalized University Recommendation Engine capable of generating tailored suggestions and providing a quantified merit prediction score for students.
- An AI-powered Guidance and Wellness Chatbot (using Google Gemini) designed to offer real-time academic advice and essential emotional/mental support to users.
- An Interactive Community Module that successfully connects students, verified alumni, and university representatives through posting, commenting, and secure one-on-one communication.
- A dedicated Admin Dashboard for university representatives to manage their institutional data, post official admission updates, and monitor student interest metrics.

2.2.2 Documentation

Comprehensive documentation is a vital deliverable that ensures the system's longevity, usability, and maintainability.

- **User Documentation:** This includes an intuitive, in-app guide and a downloadable user manual. It explains how a student can set up their profile, navigate the different modules (e.g., community, recommendations), interpret the merit prediction results, and interact effectively with the AI chatbot and alumni network.
- **Technical Documentation:** This addresses the project's structure and implementation details. It includes the Flutter project structure, the widget hierarchy, detailed explanations of backend integration with Firebase, the specific APIs used (for AI services), and the chosen state management methods (e.g., Provider or Bloc). It also covers platform-specific configurations for both iOS and Android builds.
- **Admin Guide:** A concise guide is provided for system administrators and university representatives, detailing how to use the admin features, including managing university data, posting official notices, and monitoring platform analytics and system health.

2.2.3 Testing Reports

The testing phase produces detailed reports as a key deliverable, verifying the system's quality and adherence to requirements. These reports include:

- Detailed results from Unit Testing for individual modules (e.g., authentication service, individual AI functions).
- Results from Integration Testing of critical, interconnected modules (e.g., chat functionality integrated with real-time notifications, or the recommendation engine integrated with the profile data).
- Results from UAT conducted with a sample of target students on both Android and iOS devices. Automated test scripts using the Flutter testing framework are also included to ensure consistent application behaviour across devices and future updates.

2.2.4 Deployment Setup

A complete and verifiable deployment setup is a final deliverable, ensuring the application is ready for public use. This includes:

- Preparation of final, secure release builds and the necessary signing of application files (APK/IPA).

- Configuration of the Firebase backend for production use, including Firebase Cloud Messaging (FCM) for push notifications and the final setup of authentication and analytics services.
- A detailed Deployment Guide outlining the steps for deploying the application to the Google Play Store and Apple App Store, including procedures for future updates, maintenance, and potential rollback.

These requirements define the necessary technological ecosystem, standards, and practices used throughout the Lumina project development.

2.2.5 Technologies and Tools

The project relied on a modern, robust, and scalable technology stack:

- **Frontend (Mobile App):** Built entirely with Flutter using the Dart programming language. This choice ensures a consistent and high-performance experience across Android and iOS platforms. The UI is implemented using the respective design standards (Material Design for Android, Cupertino-style widgets where needed for iOS) to be fully responsive.
- **Backend Services:** The entire backend is hosted on Firebase, utilizing:
    - **Firebase Authentication** for secure user login.
    - **Firestore** for scalable, structured storage of university data, user profiles, and community posts.
    - **Firebase Realtime Database** for low-latency, real-time synchronization of chat messages and immediate notifications.
- **AI Services: Google Gemini API** (or similar dedicated APIs) is used for building the intelligent core, including the AI chatbot, powering the recommendation logic, and facilitating data analysis for merit prediction.

2.2.6 Performance and Scalability

Lumina is engineered to handle large volumes of data and a growing user base without performance degradation.

- **Performance Optimization:** Flutter inherently ensures near-native performance. Optimization techniques implemented include efficient widget tree management, lazy

loading of images in the community feed, and asynchronous data handling to prevent UI freezes during heavy data retrieval.

- **Scalability:** By utilizing the serverless architecture of Firebase, the application's backend services (database, authentication, and cloud functions) are scalable by design, automatically handling user growth and expansion of the university data catalog without requiring continuous, manual infrastructure overhead.

2.2.7 Security and Privacy

Security is a paramount requirement, especially given the sensitive nature of student data and academic scores.

- **Secure Authentication: Firebase Authentication** is used, ensuring password hashing and secure credential storage, complemented by Role-Based Access Control (RBAC) to restrict user actions based on their role (Student, Alumnus, University Rep).

- **Data Protection:** All API communication is secured via HTTPS/SSL/TLS encryption. User data stored in Firebase is encrypted, and the application design maintains compliance with international mobile privacy standards (e.g., Google Play Store and Apple App Store guidelines).

2.2.8 Collaboration and Version Control

The development process utilized industry best practices for team collaboration and source code management. Git was used with a platform like GitHub/GitLab for Source Code Management (SCM) and Version Control. A defined branching strategy (e.g., separate branches for main, development, and individual feature branches) ensured organized, trackable, and safe development, allowing multiple team members to work concurrently.

2.2.9 Development Milestones

The project was structured across the four defined Iterative/Agile phases to ensure a systematic and efficient progression. (Referencing the Methodology section: Foundation Development, Core Functionalities, AI Features and Enhancements, and Optimization & Advanced Features). These milestones ensured that dependencies were resolved early and the system evolved logically from a stable foundation to a fully featured intelligent platform.

2.3 Overall Description

2.3.1 Product Perspective

Lumina is a new and innovative career counselling platform designed to overcome the limitations of traditional university guidance websites. Existing solutions like EduVision and IlmKiDuniya provide only static information about university admissions, lacking personalized recommendations, AI-driven merit predictions, and interactive community engagement. Lumina bridges this gap by offering a feature-rich, AI-powered platform that combines university recommendations, a real-time merit prediction model, an AI chatbot for career guidance, and a student community for peer discussions all in one place. Unlike generic career counselling services, which often rely on manual consultation and outdated data, Lumina leverages machine learning algorithms and real-time university updates to provide students with accurate, tailored advice. With an intuitive Flutter-based cross-platform experience, Firebase-powered real-time notifications, and a seamless onboarding process, Lumina ensures that students never miss opportunities and make well-informed academic decisions with ease.

2.3.2 Operating Environment:

Lumina will be developed as a cross-platform mobile application using Flutter, ensuring seamless functionality on Android and iOS devices. The backend will be powered by Firebase, providing real-time data synchronization, authentication, and cloud storage. For machine learning-based merit prediction, frameworks like PyTorch [6] will be used. The app will rely on a stable internet connection for real-time updates, chatbot interactions, and notifications. Additionally, a Flutter Web version will be implemented for university administration, allowing them to post updates and manage their profiles. The system will be optimized for smartphones and tablets, ensuring a smooth user experience across different devices.

2.3.3 Design and Implementation Constraints

Design and implementation constraints are critical in defining the boundaries and feasibility of the Lumina project. These constraints ensure that the platform remains efficient, scalable, and user-friendly while delivering high-quality career counseling and university recommendations. Below are the key design and implementation constraints that could impact the development of Lumina.

2.3.3.1 User Interface (UI/UX)

The user interface must be intuitive, engaging, and responsive, ensuring that students and universities can easily navigate the platform. The UI should allow students to input their academic details seamlessly, receive recommendations, and interact with the community. The design should support both mobile (Android/iOS) and web versions while maintaining consistency in user experience across different screen sizes.

2.3.3.2 Data Management

Efficient data management is essential for handling student profiles, university details, and AI-based merit predictions. The system must ensure fast and secure data retrieval while maintaining data integrity. University information, admission deadlines, and merit trends should be regularly updated using web scraping and API integration.

2.3.3.3 Security

Since Lumina handles sensitive student data, such as academic records and financial information, security is a top priority. The platform must implement secure authentication (Firebase Auth), encrypted data storage, and access control measures to protect user information. Compliance with data privacy regulations will be necessary to ensure student data is handled responsibly.

2.3.3.4 Integration

Lumina will integrate third-party services, such as Firebase for backend management, PyTorch for AI-based merit prediction, and web scraping tools for university data collection. The system must also ensure smooth integration of real-time notifications and AI chatbot functionalities to enhance user experience.

2.3.3.5 Compatibility

The application must be cross-platform and work seamlessly on Android, iOS, and web browsers. It should be optimized for various screen resolutions and operating systems to provide accessibility to a wide audience. The web version will primarily serve universities and admin users, ensuring easy management and content updates.

2.3.3.6 Cost Constraints

Budget limitations will influence the choice of technologies and cloud services used in the development. Cost-effective solutions like Firebase (free tier for authentication and database), open-source AI models, and Flutter for cross-platform development will be prioritized. The goal is to maintain affordability without compromising functionality or security.

2.3.3.7 Time-to-Market

Lumina is designed to address an urgent gap in career counseling, so timely development and deployment are crucial. The project will follow an Agile development methodology [11], with iterative releases and continuous feedback integration to ensure a feature-complete and stable version is launched within the given timeframe.

2.4 Current System:

Existing academic guidance platforms such as EduVision and IlmKiDuniya provide valuable information related to universities, degree programs, scholarships, and admission announcements. However, these platforms operate primarily as web portals and lack optimization for mobile use, making them less accessible for students relying on smartphones. The user interfaces often contain excessive advertisements and cluttered layouts that hinder smooth navigation and quick access to information. Another critical gap is the absence of mental health or emotional support tools. Considering the rising stress levels among students due to academic pressure, the lack of psychological support or therapy guidance is a major limitation.

Overall, while current systems provide static academic information, they fall short in usability, modern design, personalization, intelligence, and holistic student support highlighting the clear need for a solution like Lumina. Moreover, the current systems do not offer personalization every user receives the same generic information regardless of their academic background, interests, or merit score. There is no AI-driven mechanism for predicting admission chances, matching students to suitable degree programs, or providing tailored guidance.

However, these platforms have several limitations:

- They are primarily web-based and not optimized for mobile devices.
- Interfaces are often cluttered with advertisements and complex navigation, reducing usability.
- They lack personalized recommendations or AI-driven decision support.
- Mental health and emotional support features are absent, limiting holistic guidance.

- Students need to manually track deadlines and merit trends, making the process time-consuming.

The following figures are the sample figures as **Figure 2.1 Edu-Vision Website** is characterized by its focus on comprehensive academic utilities, including the distribution of date sheets, academic results, admission guides, and past examination papers across various educational boards. In contrast, and **Figure 2.2: Ilm Ki Duniya Website** emphasizes career counselling and aptitude assessment services, clearly visible through its interface promoting career choices and admission information. The inclusion of both sources highlights a cross-platform approach to research, integrating detailed academic support (IlmKiDunya) with professional guidance and career pathing tools (EduVision) to ensure a well-rounded and context-specific data foundation for Lumina.

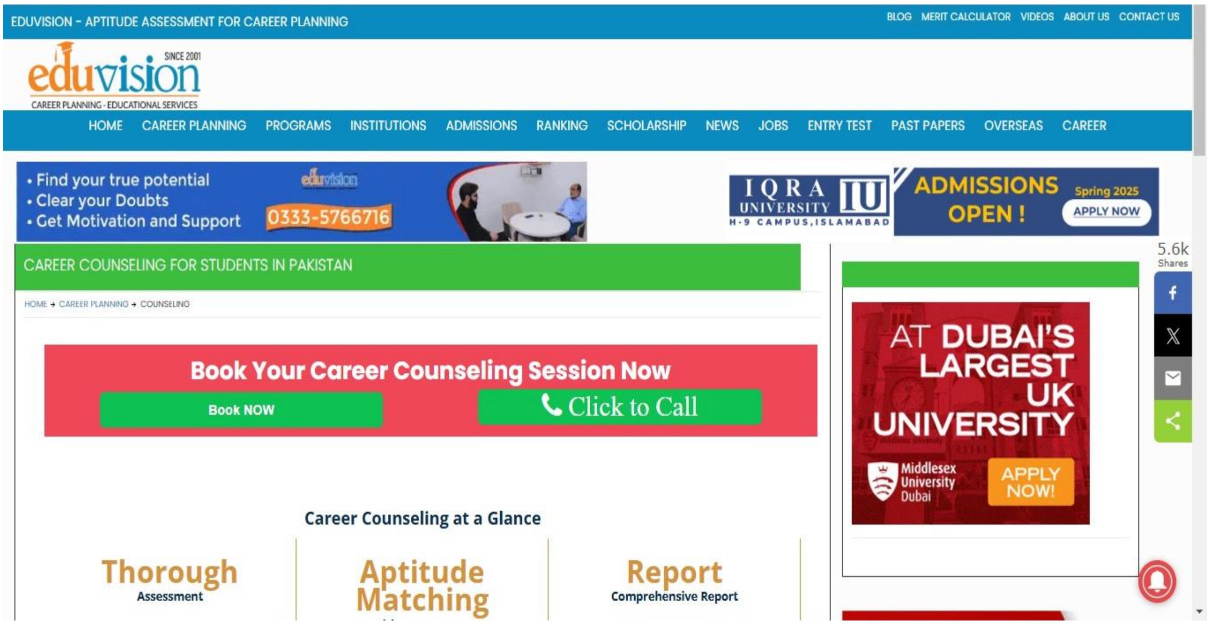

Figure 2.1 Edu-Vision Website

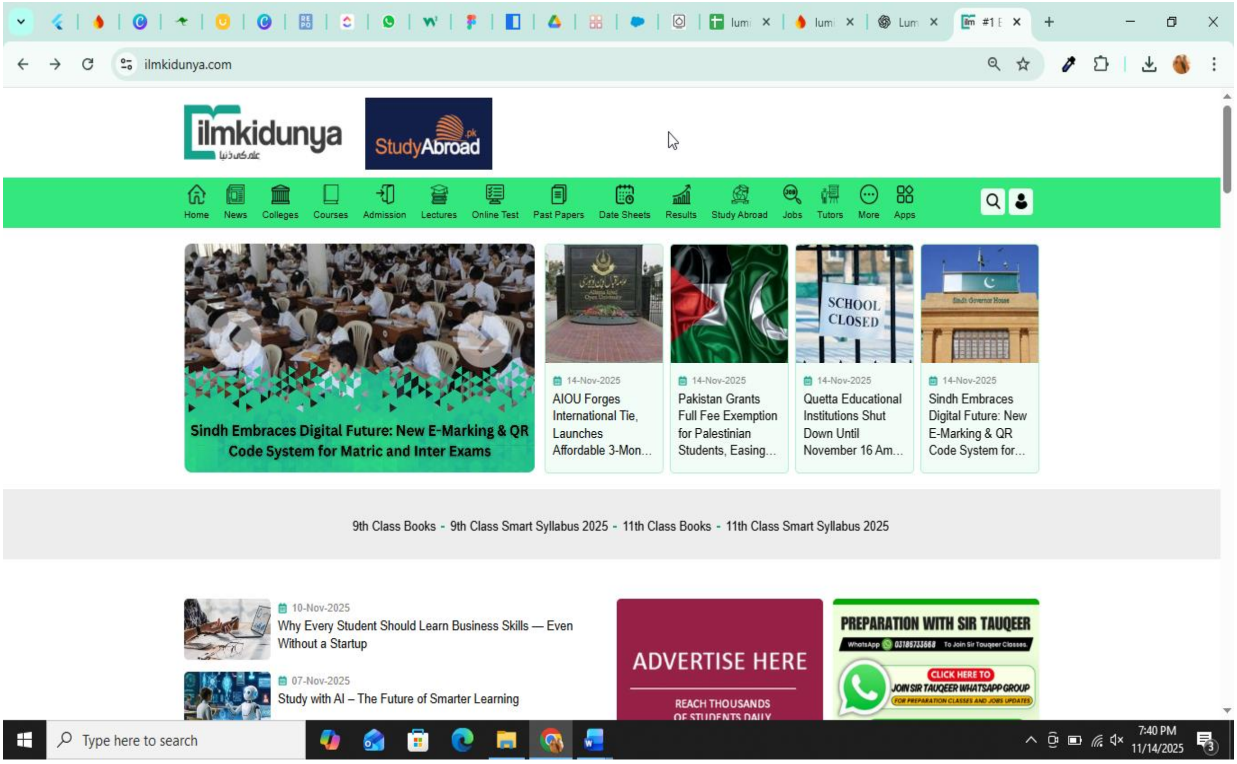

Figure 2.2: Ilm Ki Duniya Website

Chapter 3

REQUIREMENT ANALYSIS

3. Requirement Identifying Technique:

Requirement analysis is the process of understanding, collecting, and refining the needs of all users involved in the Lumina system students, alumni, and university administrators. The purpose of this phase is to identify what functionalities the system must provide, what constraints it must operate under, and how each user will interact with the application. Through stakeholder interviews, surveys, task analysis, and feedback sessions, we determine the core goals of the platform, such as university recommendations, merit prediction, community interaction, and administrative management. This analysis helps translate user expectations into clear, structured functional and non-functional requirements. The resulting use case diagram represents these requirements visually by illustrating the interactions between different user roles and the key system functionalities they need.

3.1 Requirement Analysis

The requirement analysis phase for Lumina, the AI-powered career counselling and university guidance application, focuses on understanding the needs of its three primary user groups: students, alumni, and university administrators. Through interviews, task analysis, and user feedback, we identified the essential features that users expect from the system, such as AI-driven university recommendations, accurate merit prediction, community interaction, real-time admission updates, and administrative management tools. This phase ensures that all functional and non-functional requirements are clearly defined before development begins. The analysis also helps determine how users will interact with Lumina's modules, ensuring a smooth, intuitive, and efficient experience. The following use case diagram visually represents these interactions and outlines the core operations each user role can perform within the Lumina platform.

Figure 3.1 Use Case Diagram of Lumina is a UML Use Case Diagram for the LUMINA system, illustrating the functional requirements and interactions between three primary actors: Student, University, and Admin. It shows the student's role in profile management, getting university recommendations, and using community features, while the University and Admin actors manage content, user data, and system verification tasks.

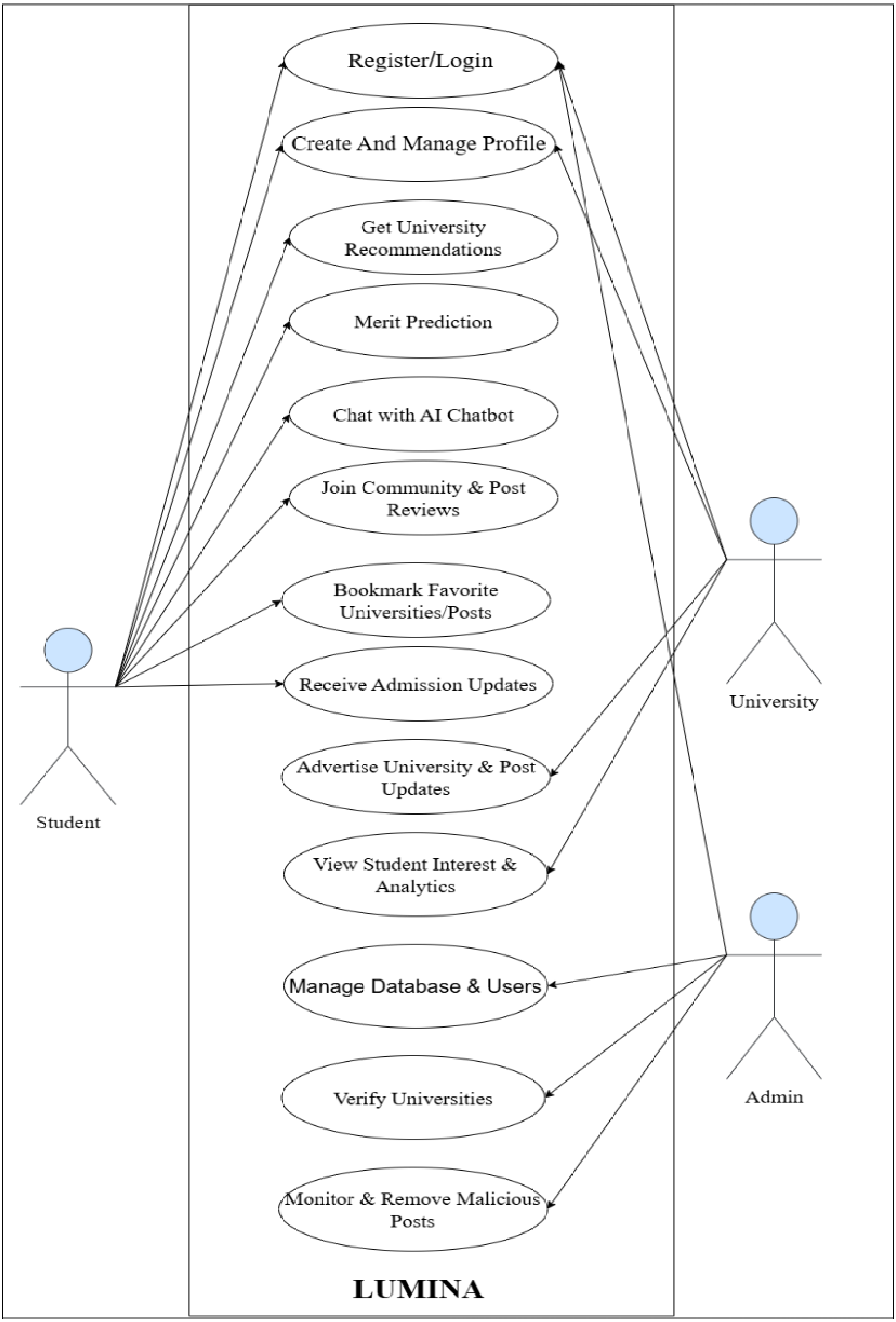

Figure 3.1 Use Case Diagram of Lumina

3.2 Use Case Description

The table below indicates a comprehensive use case template filled in with anexample drawn from the Lumina Application.

3.2.1. Register/Login (UC ID 01):

From **Table 3.1 Registration**, the Registration use case allows all major actors Students, Universities, and Admins to securely access the Lumina platform. It verifies credentials against the database for login or creates a new user record upon successful registration. The process ensures only authenticated users can access personalized and protected features of the system.

Table 3.1 Registration

Use case id	01
Use case name	Register/Login
Actors	User University Admin
Description	Users (Admin, students and universities) register or log into the Lumina platform.
Trigger	User requests to log in or register.
Pre-condition	The user has an account (for login).
Postcondition	The actor is now logged into the system, or the user's details are stored in the database (for registration).
Normal flow	The user selects Register/Login on the home screen. The system prompts the user to enter email/phone number and password. The user enters the required details. The system validates the credentials. If valid, the user is logged in successfully. If registering, the system creates a new user profile and stores it in the database.
Alternative flow	If the entered credentials are incorrect, an error message is displayed. The user is prompted to re-enter the correct details.
Exceptions	Incorrect credentials. Request to re-enter the login details.
Business rules	The system must be connected to the network.

3.2.2 Create and Manage Profile (UC ID 02):

This use case **Table 3.2 Creating & Managing Profile** enables users, specifically Students and University Representatives, to establish and maintain their presence on Lumina. It allows users to input critical data, such as academic scores, personal preferences, or official university details. A complete and updated profile is a pre-condition for personalized services like university recommendations.

Table 3.2 Creating & Managing Profile

Use case id	02
Use case name	Create and Manage Profile
Actors	Student User University Representative
Description	Users can create and update their profiles, including personal details and preferences.
Trigger	User selects the profile section to create or update details.
Pre-condition	User's must be logged into the system.
Postcondition	User's updated profile information is stored successfully.
Normal flow	The user selects "Profile" from the menu. The system displays the profile details. The user edits or adds information (e.g., name, contact, bio, preferences). The system saves the changes.
Alternative flow	The user is not logged in: Redirect to login page.
Exceptions	If the network is unavailable, profile updates are not saved.
Business rules	Profile details must be valid and follow the system's guidelines.

3.2.3 Get University Recommendations (UC ID 03)

This is a core feature explained in **Table 3.3 University Recommendation** where the system analyzes a student's profile, academic records, and stated preferences. The output is a personalized, filtered list of universities and programs that align with the student's background. This functionality helps students efficiently identify suitable academic pathways without manual, scattered research.

Table 3.3 University Recommendation

Use case id	03
Use case name	Get University Recommendations
Actors	Student User
Description	The system provides personalized university recommendations based on user preferences.
Trigger	User requests university recommendations.
Pre-condition	Users must have a completed profile with academic details.
Postcondition	Users receive a list of recommended universities.
Normal flow	The student selects Get Recommendations. The system analyzes the student's profile and academic details. The system generates a list of recommended universities. The student views and explores the recommendations.
Alternative flow	If the student profile is incomplete, the system prompts them to add details.
Exceptions	No suitable recommendations available.
Business rules	Recommendations are based on merit, preferences, and location.

3.2.4 Merit Prediction (UC ID 04)

Leveraging AI, this use case allows the student to input their current academic scores (e.g., GPA or test scores) for a probabilistic assessment of their admission chances. **Table 3.4 Merit Prediction** is a detailed specification of the Merit Prediction use case (ID 04), outlining the steps for a Student User to receive an estimated admission probability based on their academic scores and historical data. The system analyses historical merit cutoffs to generate an estimated probability of success. This prediction helps students make confident application decisions and manage expectations.

Table 3.4 Merit Prediction

Use case id	04
Use case name	Merit Prediction
Actors	Student User.
Description	The system predicts the student's admission chances based on their academic scores.
Trigger	Student requests merit prediction.

Precondition	User must provide academic scores.
Postcondition	User receives an estimated probability of admission.
Normal flow	The student selects Merit Prediction. The system requests academic details (e.g., GPA, test scores). The user enters their academic details. The system analyzes the data and generates a probability score. The student views the prediction results.
Alternative flow	If the user enters incorrect data, an error is displayed.
Exceptions	If the system lacks sufficient data, predictions may not be generated.
Business rules	Predictions are based on historical admission data.

3.2.5 Chat with AI Chatbot (UC ID 05)

Students use this feature for instant, real-time guidance and support. The AI chatbot described in **Table 3.5 AI Chatbot**, integrated with Lumina's knowledge base, addresses admissions queries, general academic advice, or provides empathetic wellness support. This module ensures immediate assistance and improves overall user experience by reducing wait times for information.

Table 3.5 AI Chatbot

Use case id	05
Use case name	Chat with AI Chatbot
Actors	Student User
Description	Students can interact with an AI chatbot for guidance on university admissions.
Trigger	User initiates a chat with the AI chatbot.
Pre-condition	The user is logged in.
Postcondition	Users receive information from the chatbot.
Normal flow	The student selects Chat with AI Chatbot. The chatbot greets the user and offers assistance. The student asks questions. The chatbot provides relevant responses.
Alternative flow	The user is not logged in: Redirect to login page.
Exceptions	AI system error: Display error message.

Business rules	The chatbot is trained on university admission-related queries.

3.2.6 Join Community & Post Reviews (UC ID 06)

This use case **Table 3.6 Community/Post Reviews** fosters social learning and peer support by allowing Students to access a dedicated community space. Students can post questions, share experiences, and write reviews about institutions or programs. It provides a structured, platform-verified alternative to external, unreliable social media forums.

Table 3.6 Community/Post Reviews

Use case id	06
Use case name	Join Community & Post Reviews
Actors	Student User
Description	Students can join a community, post reviews, and interact with other students.
Trigger	User navigates to the community section.
Pre-condition	The user must be logged.
Postcondition	User posts a review or joins discussions.
Normal flow	The student navigates to the Community Section. The student views existing posts and discussions. The student writes and submits a review.
Alternative flow	The user is not logged in: Redirect to login page.
Exceptions	Server error: Display error message.
Business rules	Community guidelines must be followed.

3.2.7 Advertise University & Post Updates (UC ID 07)

This feature is exclusive to verified University Representatives, allowing them to communicate directly with prospective students.

Table 3.7 University Advertisement details the University Advertisement and Post Updates use case (ID 07), which allows a University Representative to submit and publish advertisements and updates to be viewed by students. They can post official admission notices, deadline changes, new programs, or institutional advertisements. This ensures students receive direct, accurate, and real-time information from the source.

Table 3.7 University Advertisement

Use case id	07

Use case name	Advertise University & Post Updates
Actors	University Representative
Description	Universities can advertise themselves and post admission updates.
Trigger	University user submits an advertisement or update.
Pre-condition	The user must be registered as a university representative.
Postcondition	The post is published for students to view.
Normal flow	The university user selects Advertise University. The system requests post details. The university user submits the post. The system publishes the post.
Alternative flow	If the post violates policies, it is rejected.
Exceptions	If the network is unavailable, the post is not submitted.
Business rules	Only university representatives can post advertisements.

3.2.8 Receive Admission Updates (UC ID 08)

This passive but critical use case ensures Students are immediately informed about new institutional posts. **Table 3.8 Admission Updates** details the Receive Admission Updates use case (ID 08), outlining how a Student User who has followed or bookmarked a university is notified of university-posted admission updates, deadlines, and changes.

Table 3.8 Admission Updates

Use case id	08
Use case name	Receive Admission Updates
Actors	Student User
Description	Students receive updates regarding university admissions, deadlines, and changes.
Trigger	A university posts an update, or a scheduled notification is sent.
Pre-condition	The student must have followed or bookmarked a university reviews.
Postcondition	The student is notified of the admission update.
Normal flow	A university posts an admission update. The system sends a notification to all interested students. The student receives the update in their notifications section. The student clicks on the notification to view the full update.
Alternative flow	The user is not logged in: Redirect to login page.

Exceptions	If the student has disabled notifications, no alert is received. If the system fails to send the update due to an error, it retries later.
Business rules	Universities can only post verified admission updates.

3.2.9 Bookmark Favorites / Saved Items (UC ID 09)

This functionality described in **Table 3.9 Bookmark** provides students with a personal organization tool to save items of interest. Students can bookmark recommended universities, important admission posts, or useful community discussions for quick retrieval later. It centralizes personalized resources, making the research process manageable.

Table 3.9 Bookmark

Use case id	09
Use case name	Bookmark Favourites / Saved Items
Actors	Student User
Description	Students can bookmark universities, posts, or recommendations for later reference.
Trigger	The student selects the **"Save"** or **"Bookmark"** option.
Pre-condition	The student must be logged in.
Postcondition	The selected item is added to the saved list.
Normal flow	The student views a university profile, admission update, or recommendation. The student selects the "Save" or "Bookmark" option. The system stores the item in the student's saved list. The student can later access saved items under the Favorites section.
Alternative flow	If the student tries to save an item **that is already bookmarked**, a message is displayed: **"Already saved"**.
Exceptions	If the network is down, the bookmark action is **queued** and retried later.
Business rules	Students can remove bookmarks anytime.

3.2.10 View Student Interests & Analytics (UC ID 10)

This feature empowers University Representatives by providing key insights into student engagement. Universities can view metrics such as follower count, post interaction rates, and message volume. **Table 3.10 Student Analytics** describes the View Student Interests & Analytics use case (ID 10), enabling a University Representative to access a dashboard displaying key student engagement metrics (followers, views, clicks) to inform their strategies.

Table 3.10 Student Analytics

Use case id	10
Use case name	View Student Interests & Analytics
Actors	University Representative
Description	Universities can view analytics on student interests, including the number of students following their profile and interacting with their content.
Trigger	A university representative accesses the analytics dashboard.
Pre-condition	The university must have an active profile on the system.
Postcondition	The university gains insights into student engagement.
Normal flow	The university user navigates to the Analytics Dashboard. The system displays student interest data, including: Number of followers (students interested in the university). Engagement metrics (views, clicks, messages, saves). Top-performing posts (which updates get the most attention). The university reviews the analytics to adjust their strategies.
Alternative flow	If no students have engaged with the university profile yet, the system shows **"No data available"**.
Exceptions	If the university account **has no active posts**, analytics will be limited.
Business rules	Data should be updated **daily** to reflect accurate insights.

3.2.11 Manage Database & Users (UC ID 11)

This core administrative function, performed by the admin, allows for the maintenance of system integrity and data accuracy. **Table 3.11 Admin Managing Database** outlines the Manage Database & Users use case (ID 11), detailing how the Admin actor, upon successful login, can perform critical functions such as updating database records and managing user accounts (warnings, suspensions, removals) with all actions securely logged.

Table 3.11 Admin Managing Database

Use case id	11
Use case name	Manage Database & Users
Actors	Admin
Description	Admin can manage the database, update records, and handle user-related actions such as warnings, suspensions, or removals.
Trigger	Admin selects "Manage Users" or "Modify Database" from the dashboard.

Pre-condition	Admin must be logged in with the correct privileges.
Postcondition	Changes to user accounts or the database are successfully saved.
Normal flow	Admin navigates to the user management or database section. Admin searches for a specific user or record. Admin takes necessary action (edit, warn, suspend, delete). The system updates the record in the database. The affected user (if applicable) is notified.
Alternative flow	If no students have engaged with the university profile yet, the system shows "No data available".
Exceptions	If unauthorized access is attempted, the system denies changes. If the database is temporarily down, changes are queued and retried later.
Business rules	Admin actions must be logged for security auditing. Critical deletions require a secondary confirmation.

3.2.12 Universities Verifications (UC ID 12)

As a high-priority administrative task, the admin uses this process to confirm the authenticity of institutions seeking to use the platform. **Table 3.12 Admin University Verification** describes the Universities Verifications use case (ID 12), detailing how the Admin reviews submitted documentation from a university and updates its status to "Verified" or "Rejected," with the crucial business rule that only verified universities can interact with students.

Table 3.12 Admin University Verification

Use case id	12
Use case name	Universities Verifications
Actors	Admin
Description	Admin verifies universities by reviewing submitted documents and approving or rejecting them.
Trigger	A university submits verification documents.
Pre-condition	The university must have provided all required documentation.
Postcondition	The university's status is updated to "Verified" or "Rejected," and a notification is sent.
Normal flow	Admin accesses the list of pending verification requests. Admin reviews the submitted documents. If valid, the admin marks the university as "Verified."

	If invalid, the admin marks it as "Rejected" and provides feedback. The university is notified of the decision.
Alternative flow	If a university submits incomplete documents, the admin can request additional details.
Exceptions	If verification fails due to a system error, the admin is prompted to retry.
Business rules	Only verified universities can post updates or interact with students.

3.2.13 Monitor & Remove Malicious Posts (UC ID 13)

Table 3.13 Admin Monitor Malicious Posts specifies the Monitor & Remove Malicious Posts use case (ID 13), which is triggered when a post is reported, allowing the Admin to review content for policy violations and either remove the post or issue a warning, with the action leading to potential user suspension for repeat offenses.

Table 3.13 Admin Monitor Malicious Posts

Use case id	13
Use case name	Monitor & Remove Malicious Posts
Actors	Admin
Description	Admin monitors reported or flagged posts and removes content that violates community guidelines.
Trigger	A post is reported by users or detected by an automated system.
Pre-condition	The post must exist in the system and have at least one violation flag.
Postcondition	The post is either removed, kept, or flagged for further review.
Normal flow	Admin accesses the list of reported posts. Admin reviews the content and assesses if it violates policies. If the post is harmful, the admin removes it. If the post is acceptable, the admin marks it as "Reviewed." The original poster is notified of the action taken.
Alternative flow	If the post contains mild violations, the admin can issue a warning instead of deletion.
Exceptions	If the post was wrongly flagged, the admin can restore it.
Business rules	Posts with multiple violations result in user warnings or suspensions.

3.3 Functional Requirements

Functional Requirements (FRs) define the specific actions the Lumina system must perform to meet the needs of its users (Students, Alumni, and Universities). They are organized by user module below.

3.3.1 Student Module Requirements

3.3.1.1 Registration and Authentication

The system must provide secure and accessible mechanisms for Students to register and log in using their email and password. This includes essential security features like email verification (to confirm user identity) and a reliable password recovery option, ensuring students can regain access to the platform without administrative intervention.

3.3.1.2 Profile Management

Students must be able to create and update their personal and academic profiles, entering critical details such as name, educational history, current scores, and career interests. This information is fundamental, as it serves as the necessary input data for the personalized AI-driven services, such as university recommendations and merit prediction.

3.3.1.3 Community Interaction

Students need the ability to engage actively with the platform's social ecosystem. This includes posting questions, sharing experiences, and commenting on community discussions to seek peer advice. Furthermore, students must be able to chat with alumni directly for one-on-one career counselling and mentorship.

3.3.1.4 Personalized Guidance Features

The core intelligence of Lumina is defined here: the system must deliver personalized university recommendations based on the student's preferences and profile data. Additionally, students should be able to utilize the AI to directly apply to universities through the platform, streamlining the application process.

3.3.1.5 Information and Alerts

To keep students informed and organized, the system must allow them to bookmark universities and posts for easy future reference. Crucially, students must receive admission updates from the universities they follow, along with timely notifications for new messages, posts, and saved item reminders.

3.3.2 Alumni Module Requirements

3.3.2.1 Alumni Authentication and Profile

Similar to students, Alumni must be able to register and log in securely. Once registered, they must create detailed profiles that highlight their professional value, including their university attended, current work experience, and specific areas of expertise, allowing students to select mentors based on credible credentials.

3.3.2.2 Mentorship and Contribution

Alumni serve as mentors, requiring the ability to mentor students by answering queries within the public community forums. They must also be able to proactively contribute value by posting career tips, articles, and insights. The system facilitates direct one-on-one interaction by allowing Alumni to chat with students for private career advice.

3.3.2.3 Notification System

To ensure effective mentorship, Alumni need to be instantly alerted when their guidance is sought. The system must receive notifications when students seek advice in the community or initiate a direct chat session, ensuring timely and responsive support is maintained.

3.3.3 University Module Requirements

3.3.3.1 Profile and Management

Universities must have dedicated accounts to register and manage their official profiles on Lumina. This allows them to control their institution's branding and information, ensuring program listings, admission requirements, and contact details are accurate and up-to-date for prospective applicants.

3.3.3.2 Communication and Updates

The primary communication function allows universities to post admission updates and official notices directly to students who follow them. This eliminates information lag and reduces reliance on external communication methods. Universities also require the ability to respond to student inquiries received via the platform's messaging or community tools.

3.3.3.3 Analytics and Engagement

For strategic marketing and recruitment, universities must be able to analyse student interest metrics, such as follower counts and engagement levels with their posts. They must also receive notifications when students follow their profile or interact with their content, giving them immediate feedback on their platform activity.

3.4 Non-Functional Requirements

Non-Functional Requirements (NFRs) specify the criteria that judge the operation of the system, rather than specific behaviours. They define the system's quality, performance, and constraints.

3.4.1 Usability

The system must be designed for maximum ease of use. This means implementing a user-friendly and intuitive interface that allows students of varying technical abilities to navigate complex academic information without frustration. Furthermore, the application must be accessible to users with disabilities (e.g., screen reader support) and the UI must be responsive across mobile, tablet, and potentially desktop devices.

3.4.2 Performance

Performance dictates the speed and efficiency of the application. A key requirement is that the system should load within 2 seconds on a standard network connection, preventing user abandonment. The app must also be built to be scalable, reliably supporting increasing numbers of users and data without performance loss. Crucially, chat messages and notifications must be delivered virtually in real-time to support dynamic interaction.

3.4.3 Security

Security is paramount to protect sensitive user data. All user data must be encrypted (both in transit and at rest) to prevent unauthorized access and data breaches. The application must strictly comply with relevant data protection regulations (e.g., GDPR [9], CCPA [10]). Additionally, financial transactions and user credentials must be secured using robust protocols like SSL/TLS encryption.

3.4.4 Portability

The system must be designed for flexibility across different environments. This requires the application to be compatible with multiple operating systems (iOS, Android) and various web browsers (if a web component is developed). The underlying codebase must be modular to allow for easy updates, maintenance, and future migration to new technologies or platforms.

3.4.5 Reliability

Reliability ensures the system operates dependably and accurately. The system must accurately store and retrieve user data without errors, ensuring academic profiles and university information are always correct. Every core process, from login to recommendations and chat, must be designed to be error-free and seamless, guaranteeing a consistently positive user experience, and the system must maintain high availability with minimal downtime.

Chapter 4

DESIGN & ARCHITECTURE

4. Design and Architecture

The following parts of Software Design Description (SDD) report should be included in this chapter.

4.1 System Architecture

Lumina is a comprehensive AI-powered career counselling solution designed to streamline the decision-making process for students choosing universities. **Figure 4.1: Design Architecture of Lumina** illustrates the system's structure, dividing it into a Presentation Layer (Flutter apps), an Application Layer (Backend Logic, AI Chatbot, Recommendation System, and PyTorch ML for Merit Prediction), and a Data Layer (Firebase, Cloudinary, and Web Scraped University Data).

Lumina is based on a **3-tier architecture** comprising the following layers:

4.1.1 Presentation Layer:

• Acts as the front-end interface for users.
• Built with modern web and mobile technologies (Flutter for cross-platform compatibility).
• Ensures responsive and accessible design for use on desktops, tablets, and mobile devices.

4.1.2 Application Layer:

• Serves as the system's backend, managing all business logic.
• Developed with Firebase and Node.js for rapid development, scalability, and ease of integration with AI models.
• Handles core operations like user authentication, career path recommendations, university suggestions, and communication between students, alumni, and universities.

4.1.3 Data Layer:

• Manages the storage of all project-related data, including student profiles, career preferences, university data, and admission records.
• Uses Firestore, a flexible NoSQL database, to ensure quick read/write operations and scalability.
• Implements encryption and strict access controls for data security, ensuring that user data is protected and privacy is maintained.

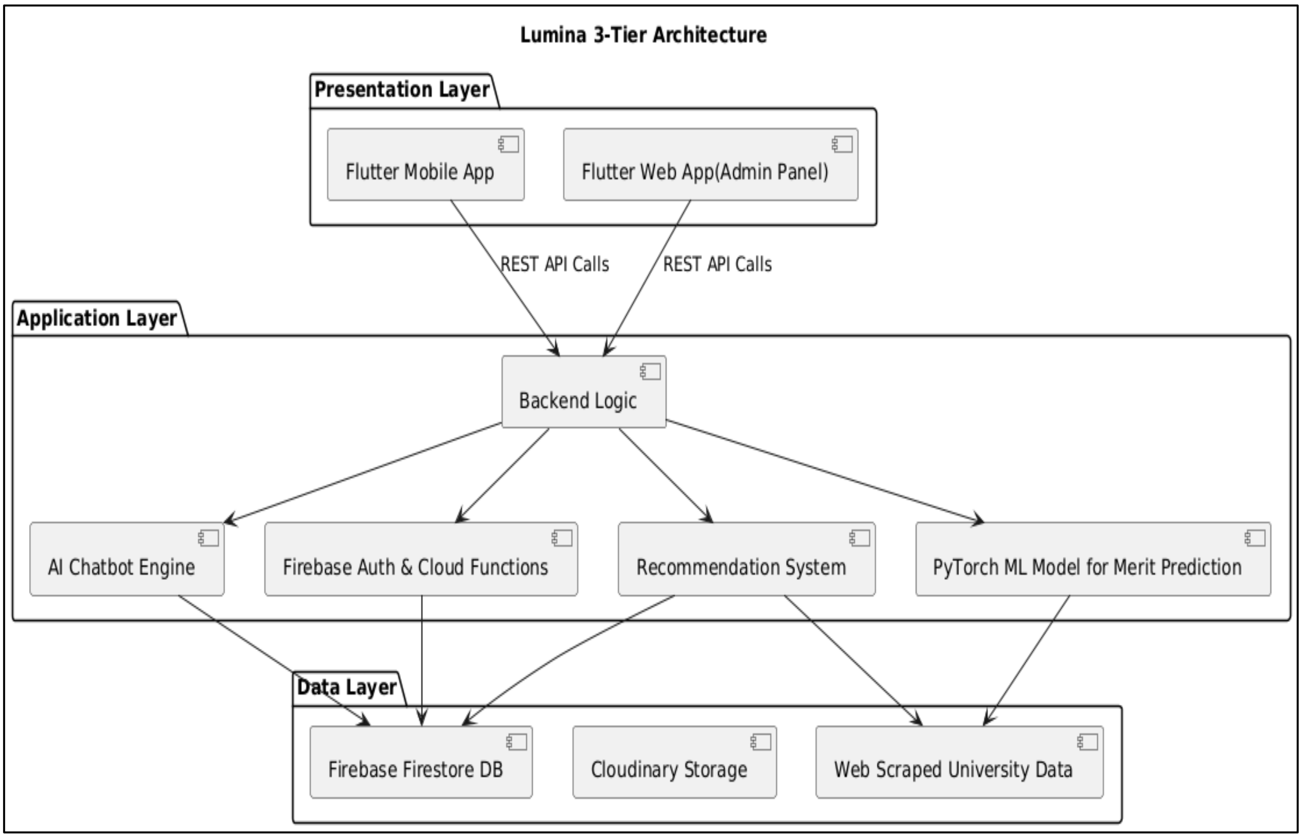

Figure 4.1: Design Architecture of Lumina

4.2 Process Flow

The Process flow diagram below in **Figure 4.*2*: Process flow of Lumina** captures the chronological flow of interactions between system components and user roles during core operations. It typically starts with a user (like a student) logging into the application, followed by interactions such as browsing career posts, initiating chat with alumni, or receiving AI-generated recommendations. The diagram represents communication between the user interface, backend services, database, and the AI engine using message arrows and activation bars. Each action, like sending a message or posting a query, is shown as a stepwise exchange that results in an appropriate system response. The sequence also includes user verification, data retrieval, and output delivery. This visual representation ensures clarity in system behavior, helps identify potential delays or bottlenecks, and validates the logic of feature implementation. It effectively models real-time interactions and backend processing for a smooth user experience.

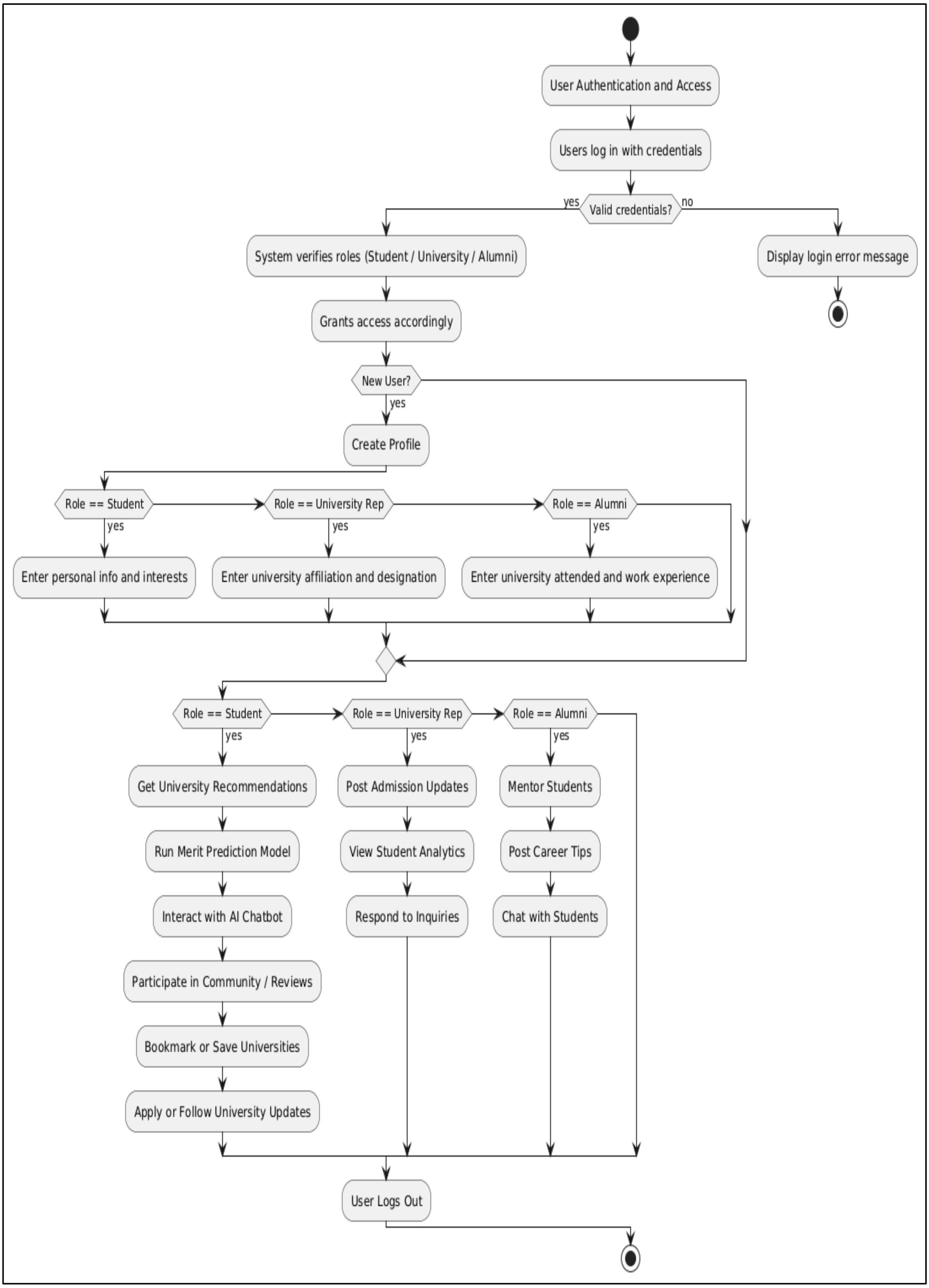

Figure 4.2: Process flow of Lumina

4.3 Design Models

This is important to include design models that provide a clear and comprehensive understanding of the system's structure and functionality.

The design model of Lumina is structured around three core components: sequence diagrams, class diagrams, and data design. The sequence diagrams illustrate dynamic interactions between users (students, alumni, universities, admins) and system modules like chat, AI chatbot, university recommender, and community forum—capturing real-time behaviours such as login, message exchange, and merit prediction. The class diagram defines the static structure of the system, modelling entities like User, Student, Alumni, University, Post, and ChatThread with their attributes, relationships, and behaviours, ensuring modularity and reusability. Finally, the data design is based on Firebase's NoSQL structure, organizing data into collections (e.g., users, posts, messages, notifications) with real-time syncing and secure access control, aligning with Lumina's cross-platform, scalable architecture. Together, these models ensure that Lumina is robust, maintainable, and optimized for interactive career counselling experiences.

4.4 Sequence Diagram

The Sequence Diagram of Lumina **Figure 4.*3*: Sequence Diagram of Lumina** captures the chronological flow of interactions between system components and user roles during core operations. It typically starts with a user (like a student) logging into the application, followed by interactions such as browsing career posts, initiating chat with alumni, or receiving AI-generated recommendations. The diagram represents communication between the user interface, backend services, database, and the AI engine using message arrows and activation bars. Each action, like sending a message or posting a query, is shown as a stepwise exchange that results in an appropriate system response. The sequence also includes user verification, data retrieval, and output delivery.

The following diagram is organized by four main user types (Actors) and seven core System Lifelines (components), showing the flow of operations from top to bottom.

4.4.1 Initial Access and Authentication

The process begins with securing user access, handled by the Auth component.

- **Login/Registration:** The User (Student, Alumni, or University) initiates the process by sending a Login/Register message to the Auth component.

- **Verification: Auth** interacts with the Profile component to Verify or Create User status.
- **Authentication Result:** Auth returns the Auth Result (success or failure) to the User.
- **Access Grant:** The final Access Granted / Denied message is sent back to the User, allowing them to proceed to the main system.

4.4.2 Profile Setup

Once authenticated, the system handles the creation or update of user profiles, crucial for role-based access.

- **Submission:** Depending on the actor, the student profile, Alumni profile, or University profile is submitted to the Profile component.
- **Saving Data:** The Profile component sends a Save profile info message to the Database.
- **Confirmation:** The Database confirms the save, and the Profile component sends a Confirmation message back to the User, indicating the Profile created/updated.

4.4.3 Student Functionalities

These sequences involve the AI, Comm., Chat, Analytics module and the Database to deliver core student services.

- **University Suggestions:**
- **Student** sends a Request University Suggestions to the Modules component.
- **Modules** fetch the user's Fetch profile from the Database to personalize the suggestions.
- **Modules** return the Recommended list to the student.
- **Merit Prediction:**
- **Student** sends a Run Merit Prediction request to Modules.
- **Modules** return the Prediction result.
- **AI Chatbot:**
- **Student** sends an Ask Chatbot (Career Q) message to Modules.
- **Modules** return the Chatbot reply.
- **Saving Data:**
- **Student** initiates a Create Post / Bookmark / Apply action.
- **Modules** execute Save post/bookmark to the Database.
- **Modules** sends Confirmed back to the student.
- **Alumni Chat Initiation:**
- **Student** sends Start Chat with Alumni to Modules.
- **Modules** interact with the Database to Fetch/Create Thread.

- **Modules** sends a Notify Alumni message to the Notification component.

4.4.4 Alumni Functionalities

This section focuses on engagement, mentorship, and communication, heavily involving the **Database** and **Modules**.

- **Responding to Posts:**
- **Alumni** send View/Reply to Student Posts to Modules.
- **Modules** sends a Save comment message to the Database.
- **Chat with Students:**
- **Alumni** send Chat with Students to Modules.
- **Modules** sends a Store message message to the Database for persistence.

4.4.5. University Functionalities

These sequences focus on data dissemination and analytics, utilizing the Notification service.

- **Posting Updates:**
- **University** sends Post Admission Update to Modules.
- **Modules** trigger the Notification component to Push to Students.
- **Viewing Analytics:**
- **University** sends View Analytics to Modules.
- **Modules** sends Get stats to the Database.
- **Modules** return Engagement insights to the University.

4.4.6 Admin Functionalities

This sequence represents administrative oversight and system maintenance.

- **Management:**
- **Admin** sends Manage Users / Posts to Modules.
- **Modules** interact with the Database to carry out the necessary operations (e.g., deletion, editing, suspension).
- **Modules** return the Success / Data confirmation to the Admin.

  This Sequence Diagram efficiently models the real-time interactions and message passing across the system's distributed components, ensuring clarity in system behavior and validating the logic of feature implementation for a smooth user experience.

  Furthermore, the Sequence Diagram acts as a validation tool for ensuring that the system's modular architecture operates cohesively. By mapping the exact order of message exchanges between components such as Auth, Profile, Modules, Database, Chat, Analytics, and Notification services, developers can confirm that no step is missing or logically inconsistent. This visualization helps identify redundant interactions, potential delays, or bottlenecks especially during high-traffic operations like messaging or simultaneous student queries. It also supports debugging by clearly outlining expected behavioral flows, which becomes critical when integrating AI-driven components that rely on continuous data retrieval and real-time responses.

Lastly, the diagram **Figure 4.*3*: Sequence Diagram of Lumina** serves as a bridge between system analysts, developers, and stakeholders by offering a clear depiction of both user-level interactions and internal system processes. Each actor's responsibilities and system privileges become transparent, improving communication and reducing ambiguity during development. This structured sequential representation ensures that the Lumina platform's implementation remains aligned with its user-centric goals—seamless access, intelligent recommendations, real-time communication, and robust administrative control. Ultimately, the Sequence Diagram not only documents how the system functions but also provides a blueprint for scalability, future enhancements, and long-term maintenance.

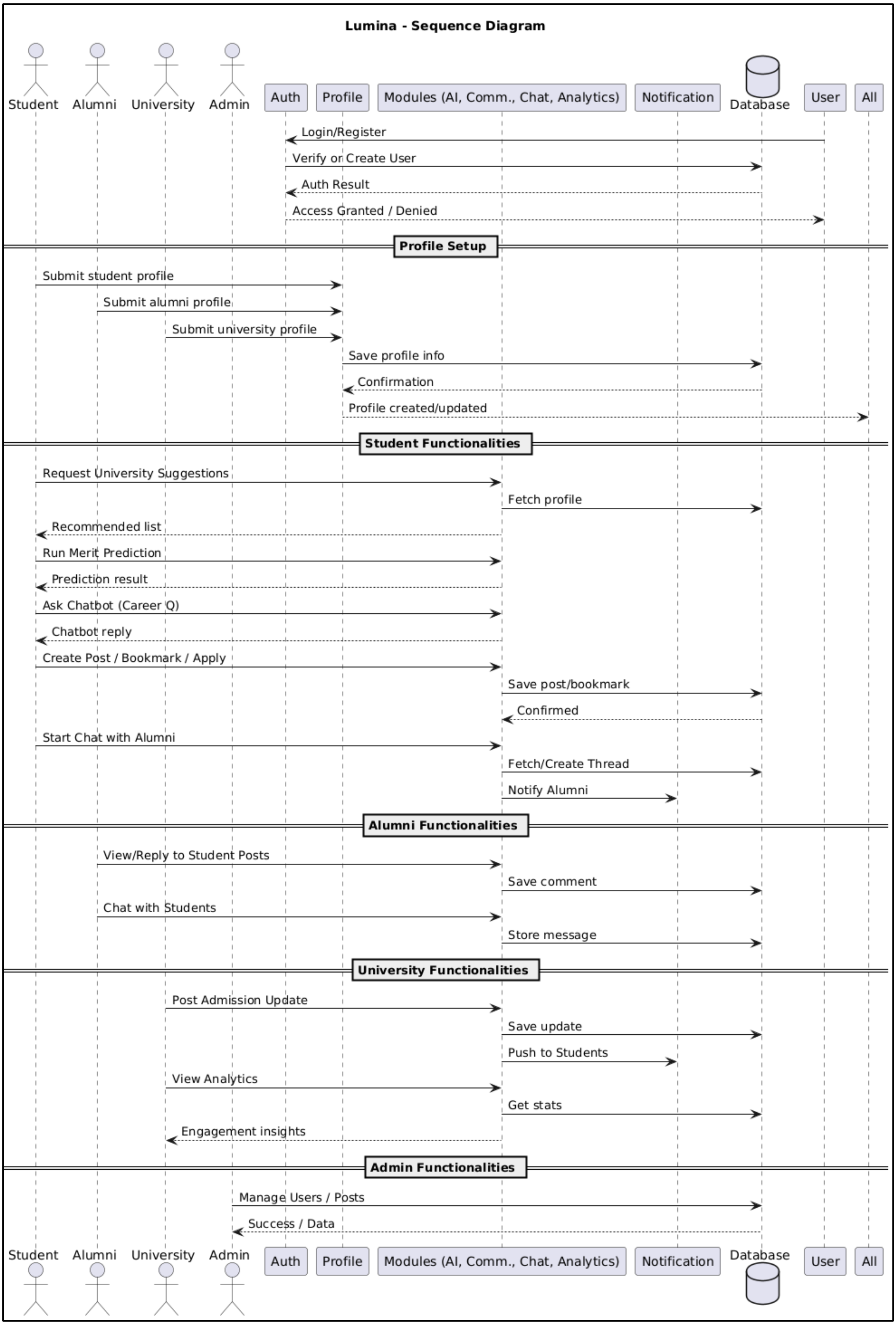

Figure 4.3: Sequence Diagram of Lumina

4.5 Class Diagram

The Class Diagram outlines the object-oriented structure of the Lumina app by displaying the main classes and their relationships. At the core is a generic User class, which is extended by specific roles such as Student, Alumni, and University. Each subclass includes attributes and methods relevant to its responsibilities, such as viewing posts, chatting, or managing notices. Other key classes include Post, Message, Chat, and AI Engine, each encapsulating specific data and functions related to content creation, communication, and intelligent suggestions. Relationships such as inheritance, associations, and aggregations clearly indicate how objects collaborate. The diagram **Figure 4.*4*: Class Diagram of Lumina** ensures data encapsulation, reuse through inheritance, and a clean modular design. It serves as a blueprint for developers to understand the app's structural layout and ensures maintainable, scalable code architecture.

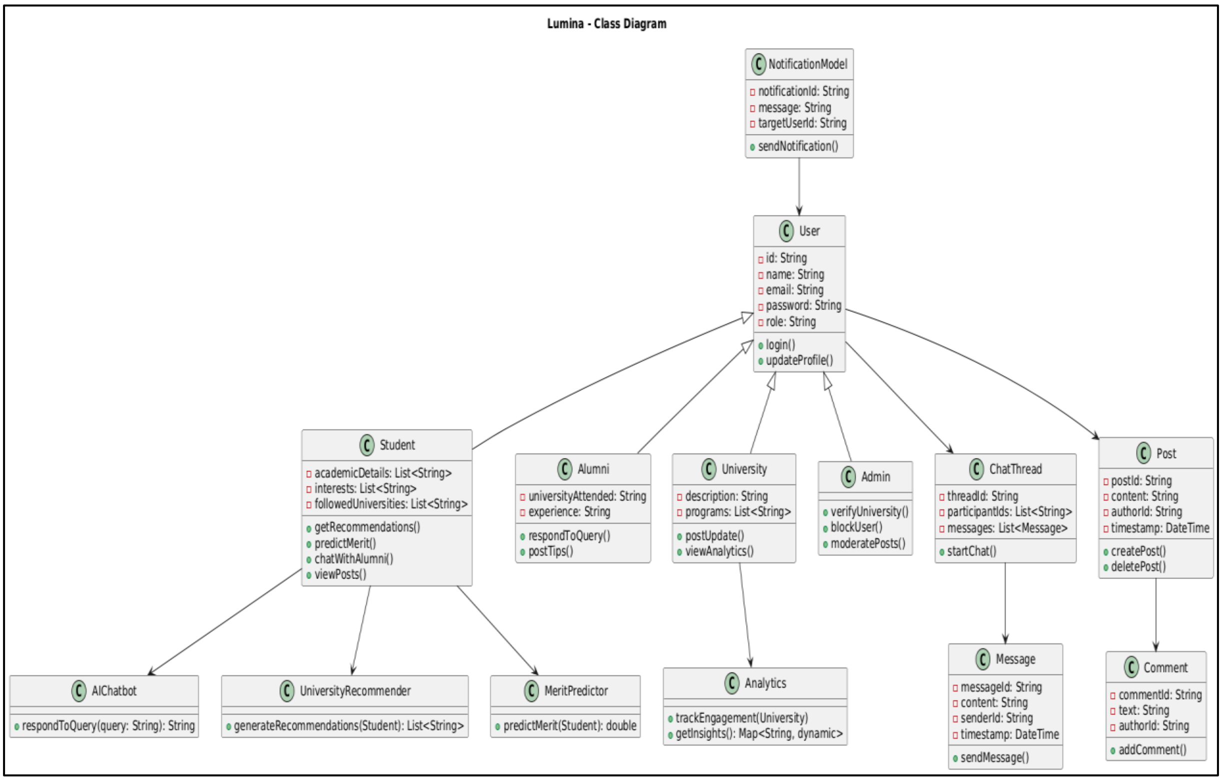

Figure 4.4: Class Diagram of Lumina

4.6 Data Design

The data design for the Lumina system consists of the key entities involved in the platform, their attributes, and the relationships among them. These entities form the foundational structure for storing and managing data, enabling operations such as user authentication, profile creation, university recommendations, career path suggestions, real-time admission updates, and communication between students, alumni, and universities.

4.7 Data Dictionary

The Data Dictionary defines the key data elements used within the Lumina system. It provides unique identifiers for each entity to ensure data integrity, consistency, and efficient retrieval across the app's modules.

- **UserID**: A unique identifier assigned to each user (student, alumni, university, or admin).
- **ProfileID**: A unique identifier for individual user profiles, storing details such as education level, interests, and career goals.
- **UniversityID**: A unique identifier representing each university listed in the system.
- **AlumniID**: A unique identifier for alumni who are available for mentoring or career advice.
- **RecommendationID**: A unique identifier for career or university recommendations generated for a user.
- **ChatID**: A unique identifier for conversations between users (student-alumni, student-university, etc.).
- **NoticeID**: A unique identifier for notices or announcements posted by universities or admins.
- **RatingID**: A unique identifier used to track feedback or ratings provided to alumni or universities by students.

Chapter 5

IMPLEMENTATION

5. Implementation

5.1 Algorithm

The algorithms implemented in Lumina form the core logic behind key intelligent and interactive features of the app. They ensure secure, scalable, and dynamic functionality across the platform. The backend is built using Firebase, integrated with AI models (e.g., for recommendations and predictions), and Flutter for the frontend.
Key algorithms include:

5.1.1 User Authentication Algorithm

This algorithm ensures secure login and controlled access based on user type (student, alumni, university admin). It is implemented using Firebase Authentication and role-based database checks.

Steps:

- **Identity Verification**
- User enters credentials (email/password or Google Login).
- Firebase Authentication verifies identity through encrypted tokens.
- **Role-Based Access Assignment**
- After successful login, the system fetches the user's role from Firestore (student, alumni, admin).
- The UI and available features are loaded according to the assigned role.
- **Session Management**
- A secure authentication token is generated and stored temporarily.
- Auto-login and logout logic ensures safe and seamless user sessions.

5.1.2 Career Recommendation Algorithm

This algorithm uses AI models and user-provided data to generate personalized suggestions for careers and universities.

Steps:

- **User Profile Data Collection**
- Inputs include academic history, skills, interests, preferred location, and financial constraints.

- **AI-Based Recommendation Generation**
- Machine learning models (e.g., LLMs via Gemini) analyze patterns and similarities with past successful student journeys.
- The algorithm scores and ranks the most suitable career paths and universities.
- **Recommendation Delivery**
- The highest-ranked results are displayed in an easy-to-understand format.
- The user can refine filters to get even more personalized outcomes.

5.1.3 Merit Prediction Algorithm

This algorithm predicts a student's chances of admission by comparing their academic performance with historical merit data from universities.

Steps:

- **Data Collection and Preprocessing**
- Uses student's marks, examination boards, and program preferences.
- Fetches historical merit lists, cutoff trends, and admission policies.
- **Model Calculation and Probability Forecasting**
- AI model calculates probability scores using regression or classification techniques.
- Predicts whether the student falls above, below, or near the merit threshold.
- **Output Interpretation**
- The system displays: "High chance," "Moderate chance," or "Low chance."
- Provides suggestions for backup universities or alternative programs if required.

5.1.4 Communication Matching Algorithm

This algorithm connects students with alumni or universities for mentorship, guidance, or clarifications.

Steps:

- **User Interest Profiling**
- Collects student interests, degree intentions, or career fields.
- Matches with alumni/university profiles that share similar academic backgrounds.
- **Compatibility Filtering**
- Algorithm filter matches through criteria such as department, expertise, location, and preferred communication type.
- **Smart Matching & Chat Initialization**

- Best match is displayed to the student first.
- Chat room is automatically created, and both parties are notified.

5.1.5 Notification & Alert Algorithm

This algorithm manages real-time alerts for admission deadlines, alumni reply, recommendations, and system updates using Firebase Cloud Messaging (FCM) [8].

Steps:

- **Trigger Event Detection**
- New admission announcement, new message received, updated merit list, or personalized suggestion available.
- **Notification Generation**
- The backend composes a notification message with the necessary details.
- Categorizes alerts (urgent, informational, personalized).
- **Delivery to User Device**
- FCM sends the notification to the appropriate user role.
- App displays notification through local channels with click actions (e.g., open chat, view announcement).

5.2 External APIs

The Lumina application integrates several external APIs shown in **Table 5.1 Details of APIs used in the project** to ensure secure authentication, scalable data storage, intelligent content generation, and smooth media handling. Firebase Authentication is used to manage user signup, login, identity verification, and session control, enabling secure role-based access for students, alumni, and universities through functions such as loginUser (), registerUser (), and getUserRole () implemented in the AuthService. The system relies on Firestore Database, a real-time NoSQL database, to store and retrieve user profiles, chat messages, alumni posts, merit lists, and recommendation data using methods like getUserData (), saveProfile (), and fetchMeritData () within DatabaseService. For optimized media storage, Cloudinary API handles profile images, university banners, and document uploads via uploadImage () and getImageUrl () managed in Media Service. Firebase Cloud Messaging (FCM) powers Lumina’s real-time alert system, sending notifications for matches, chat messages, and recommendation updates through onMessageReceived () and triggerNotification () under FCMService. Additionally, the Gemini API (Google AI Studio) enables AI-driven functionality such as personalized career recommendations, merit prediction, and profile analysis using generateRecommendation (), predictMerit (), and analyzeProfile () in the AIService. Together, these APIs create a secure, intelligent, and seamless user experience across the Lumina platform.

Table 5.1 Details of APIs used in the project

Name of API	Description of API	Purpose of usage	List down the function/class name in which it is used
Firebase Authentication	Provides secure user login, signup, identity verification, and session control.	Secure login, role-based access, and session management.	AuthService, loginUser(), registerUser(), logoutUser(), getUserRole()
Firestore Database	Real-time NoSQL cloud database by Firebase.	Stores user profiles, alumni data, recommendations, chat data, merit lists, and career guidance information.	DatabaseService, getUserData(), saveProfile(), fetchMeritData(), saveChatMessage()
Cloudinary API	Cloud-based image hosting and optimization service.	Stores and delivers optimized images for user profiles, university banners, and documents.	MediaService, uploadImage(), getImageUrl()
Firebase Cloud Messaging (FCM)	Real-time messaging and notification system.	Sends in-app alerts, chat notifications, and personalized recommendation alerts.	FCMService, onMessageReceived(), triggerNotification()

Gemini API (Google AI Studio)	LLM-based AI API for analysis, ranking, and generating chatbot responses.	Career recommendations	AIService, generateRecommendation(), predictMerit(), analyzeProfile()

5.3 User Interface

The Human Interface Design of Lumina is centered around creating an intuitive, visually cohesive, and high-performance experience tailored for students, alumni, universities, and administrators. Our approach utilizes a UCD prioritizing clarity, simplicity, and efficiency across all roles. The interface adheres to modern material design principles, employing a sleek, minimalist aesthetic characterized by generous white space, a focused teal-based color palette, and clear typographical hierarchy. This design philosophy is maintained throughout the entire application—from the initial authentication flow to complex data visualization on the Admin Panel—ensuring a seamless environment that supports real-time interaction, personalized career guidance, and intuitive platform navigation for making informed academic and professional decisions.

Beyond aesthetics, Lumina's interface is engineered to optimize usability through thoughtful interaction design and adaptive accessibility features. Each module—whether it involves university recommendations, alumni networking, merit prediction, or administrative analytics—is structured with clear task flows, consistent component behaviour, and minimal cognitive load. Dynamic dashboards, responsive layouts, and real-time feedback ensure that users can effortlessly engage with the system on any device. Additionally, smart UI enhancements such as contextual prompts, guided form inputs, and adaptive content presentation help users navigate complex actions with ease. Together, these interface decisions enhance trust, reduce friction, and strengthen the overall engagement experience, making Lumina not only visually appealing but functionally powerful for all user groups.

5.3.1 Authentication Screens

The application begins with clean, responsive Authentication Screens **Figure 5.*1*: Signup Screen** and **Figure 5.*2*: Login Screen** that serve as the gateway to the Lumina platform. These interfaces are designed for minimal friction, featuring intuitive form fields with real-time input validation to guide users smoothly through the process. A dedicated role-specific sign-in flow ensures that students, alumni, and university representatives are routed correctly upon registration. The sleek design emphasizes a clear call-to-action e.g., "Sign In" or "Register" and includes accessible features like password recovery and social login options, ensuring a smooth and secure onboarding experience. Clean and responsive login/register interfaces with input validation, password recovery, and role-specific sign-in flow.

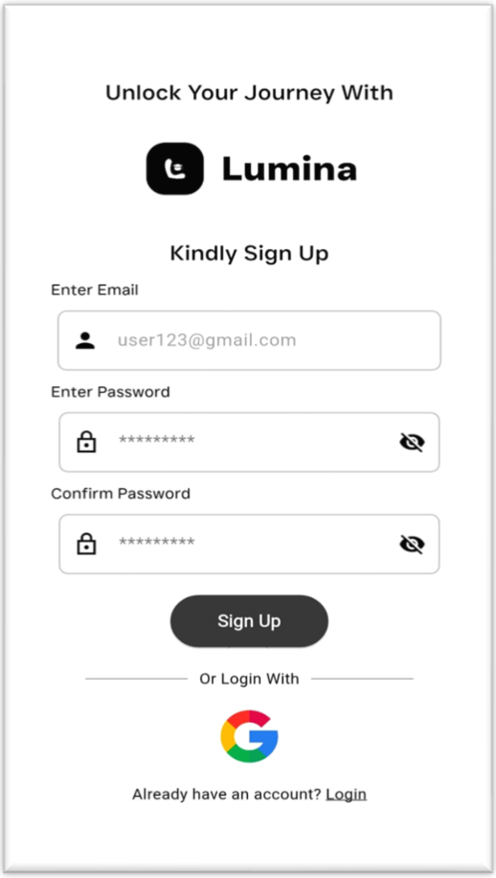

Figure 5.1: Signup Screen

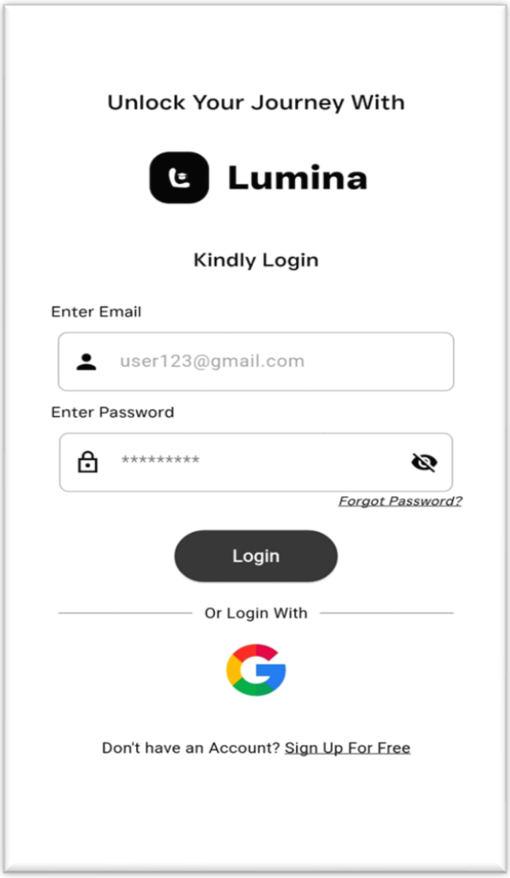

Figure 5.2: Login Screen

5.3.2 Profile Creation and Edit Profile Screens

Following authentication, users are immediately guided through a structured Profile Creation process designed to capture essential data personal, educational, and career-related—which forms the foundation of their platform identity. This process is broken down into manageable steps to prevent user fatigue. The subsequent **Figure 5.3: Edit Profile Screen** maintains a consistent, role-based layout, allowing students, alumni, and universities to easily update their bio, education, work experience, and interest-based details. The interface is visually segmented with clear section headers and utilizes modern, minimalist input fields for effortless data management and editing. **Figure 5.4: Setup Profile Screen** used for students to select their academic interests, utilized for context and data gathering.

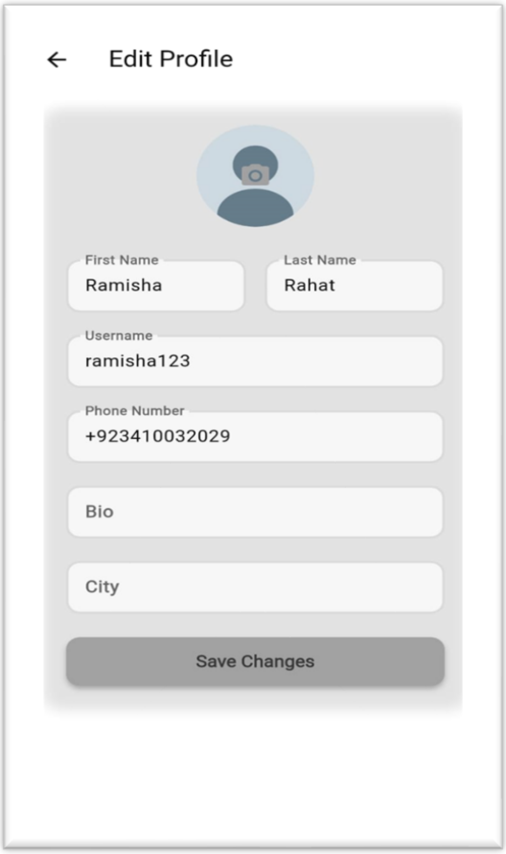

Figure 5.3: Edit Profile Screen

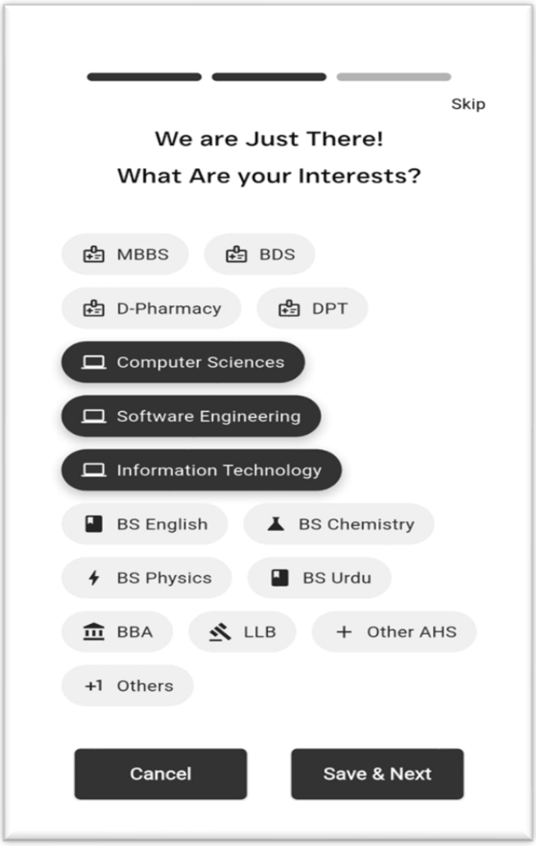

Figure 5.4: Setup Profile Screen

5.3.3 Welcome & Onboarding Screens

The Welcome Screens as in **Figure 5.5: Welcome Screen** deliver a warm, engaging introduction to Lumina's core value proposition and features, setting a positive tone for the user journey. The initial Onboarding sequence is concise and utilizes visual cues and short explanatory texts to familiarize users with the main navigation and functionalities. This initial guidance helps new users quickly understand the platform's benefits, such as searching for mentors or accessing AI assistance, ensuring a low barrier to entry and promoting immediate feature adoption. **Figure 5.6: Profile Screen** is the LUMINA Profile Screen displaying a user profile for "Ramisha Rahat" with tabs for Posts, Liked, and Bookmarks, and a bottom navigation bar for University, Community, Chatbot, Search, and Profile.

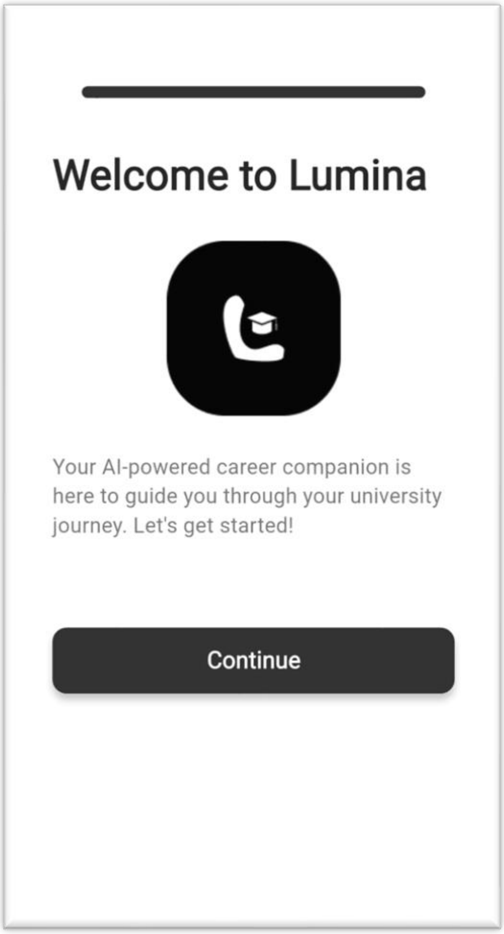

Figure 5.5: Welcome Screen

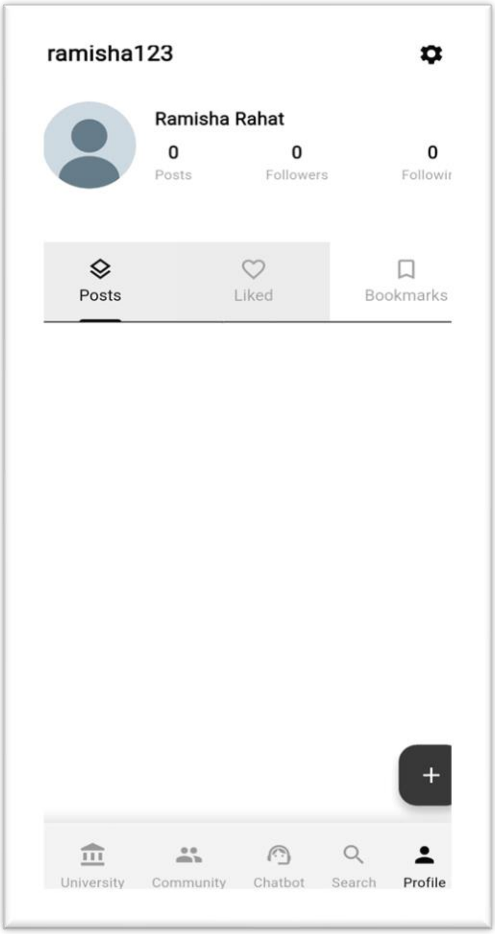

Figure 5.6: Profile Screen

5.3.4 Search Screen to Search All Users

The Search Screen Mockup provides a centralized, high-efficiency interface for discovering all platform entities universities, alumni, posts, and fellow students. **Figure 5.7: Search Screen** is the LUMINA Search Users Screen, showing the interface where a user can search for other users (like "arham1") and initiate a Follow action, with the Search icon highlighted in the bottom navigation bar. The design features a prominent, highly responsive search bar complemented by advanced filtering options (e.g., location, program, graduation year) presented in a sleek, collapsible panel.

5.3.5 Chat & Bot Screens

The Chat & Bot Screens **Figure 5.8: Chatbot Screen** facilitate two crucial modes of communication. The real-time 1:1 Chat Interface for student-alumni communication is designed with a familiar messaging layout, featuring clear message bubbles, read receipts, and media sharing support for fluid interaction. Alongside this is a dedicated, responsive AI

Chatbot Screen, offering instant, contextual career guidance through a clean, conversational interface. The focus here is on simplicity and speed, ensuring users receive timely, direct assistance without complex navigation.

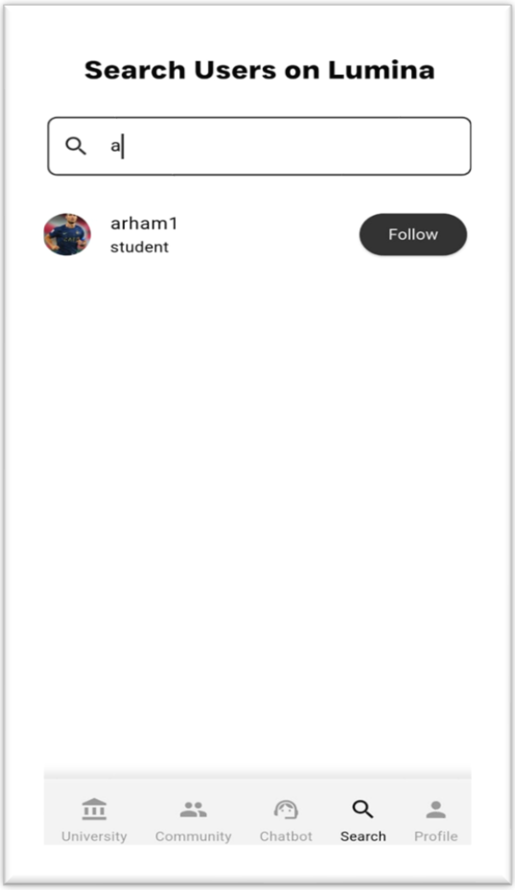

Figure 5.7: Search Screen

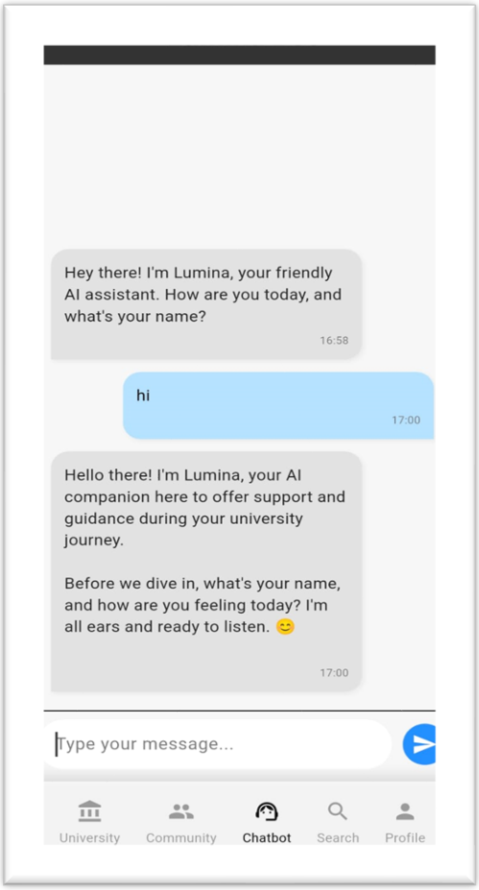

Figure 5.8: Chatbot Screen

5.3.6. Creating Post Screen

Figure 5.9: Post Uploaded Screen shows the LUMINA mobile app's "Create Post" screen, where a user uploads an image and enters the "first post" caption into the input field with a maximum limit of 220 characters. **Figure 5.10 Post Uploading Screen** is the "Create Post" screen from the LUMINA mobile application, showing a template for uploading an image (JPEG or PNG, Max 5MB) and adding a caption (Max 220 characters).

Figure 5.9: Post Uploaded Screen

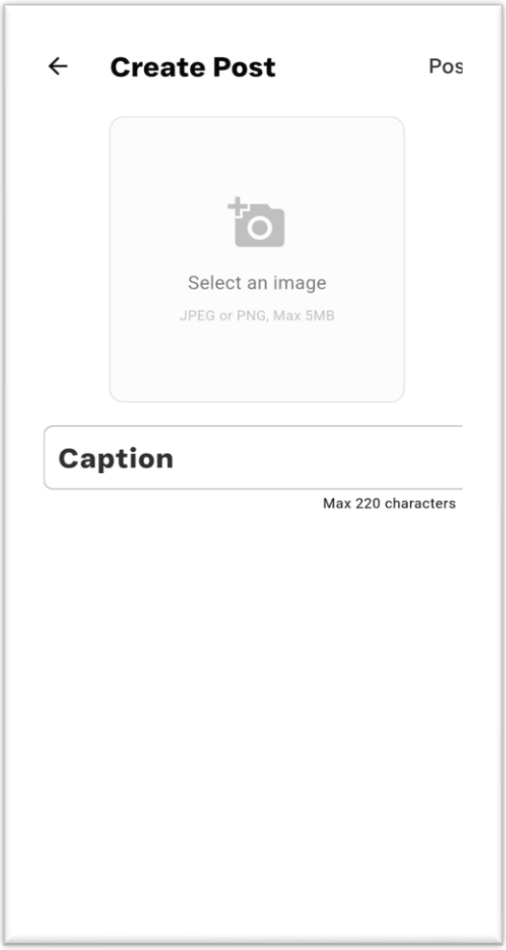

Figure 5.10 Post Uploading Screen

5.3.7. Message Inbox Screen for One-on-One and Group Conversations

The Message Inbox Screen **Figure 5.11: Inbox Screen** serves as the central hub for all user communications. It presents a clear, chronological list of active one-on-one chats and potential group conversations. The inbox utilizes a sleek, two-page layout on wider screens (or a stacked layout on mobile) to display conversation summaries on the left and the active chat thread on the right, providing an efficient overview and quick access to ongoing mentor-mentee dialogues.

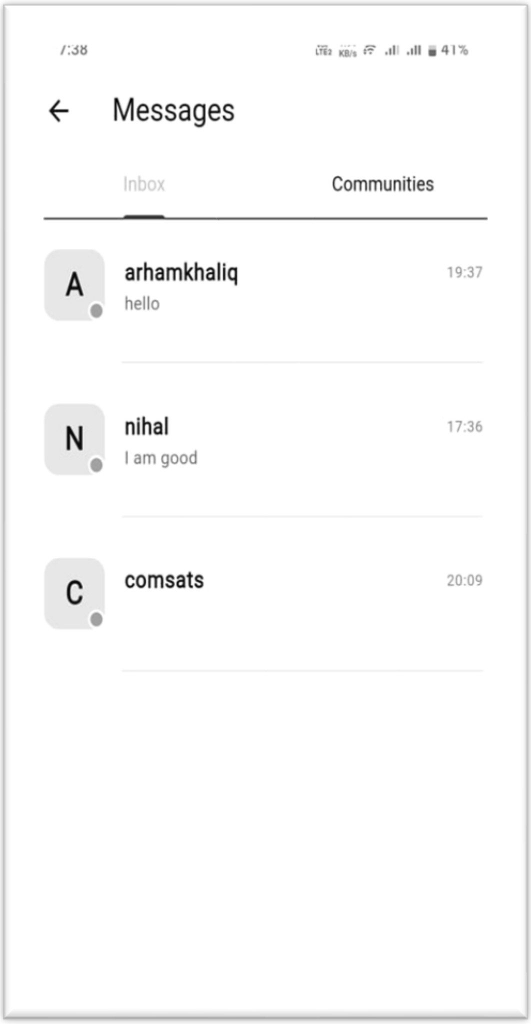

Figure 5.11: Inbox Screen

5.3.8. Admin Panel for Lumina

The Admin Panel **Figure 5.*12*: Admin Panel Dashboard** is a secure, analytics-driven interface designed for smooth platform moderation, distinguished by a clean, professional aesthetic.

- **a. Dashboard:** The Admin Dashboard utilizes visual data representation (as implemented with the sleek chart updates) to provide an immediate overview of platform health, including total user/post counts, user activity trends, and demographic breakdown, enabling data-driven decision-making.
- **b. User Registration and Verification Screen:** This dedicated screen features a highly scannable data table (with minimal borders and clear hierarchy) to manage and oversee user registration, crucially including a robust verification interface to approve or reject universities based on submitted credentials, ensuring data integrity and platform security.

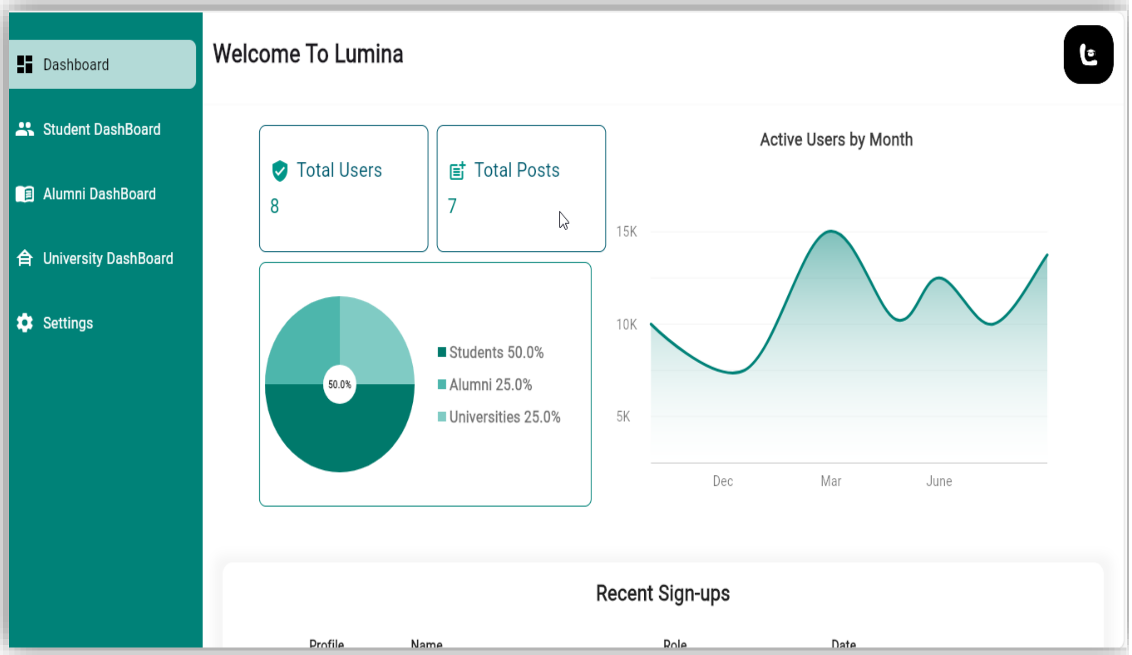

Figure 5.12: Admin Panel Dashboard

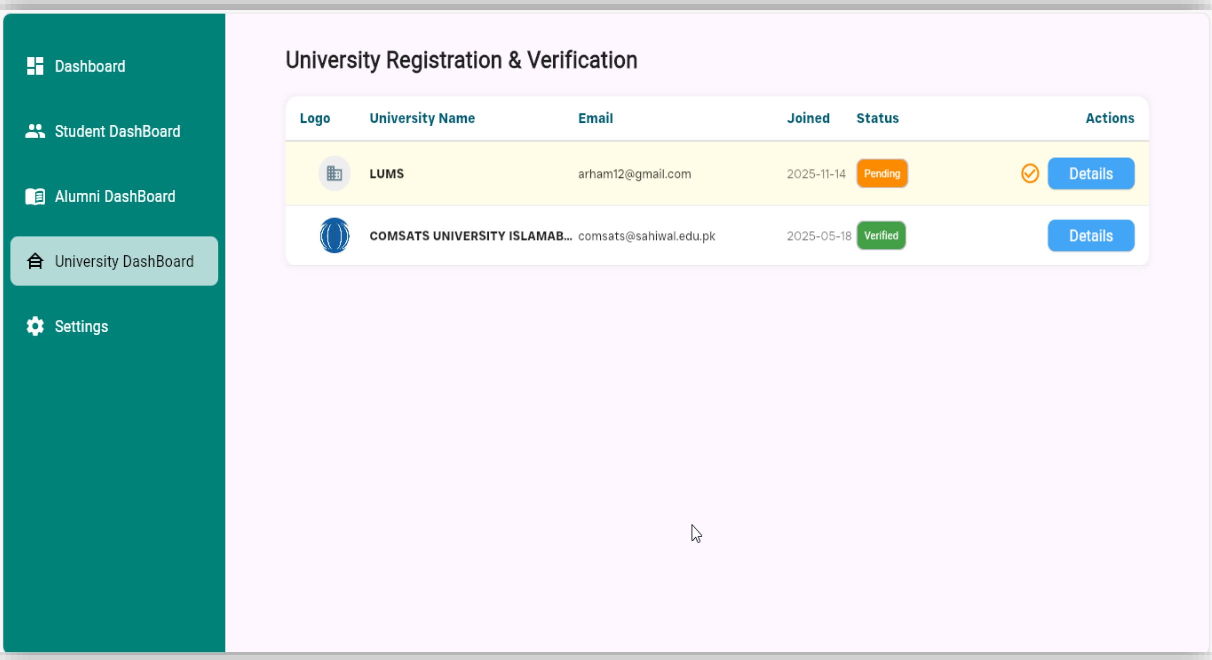

Figure 5.13: Admin Panel University Dashboard

Chapter 6

TESTING & EVALUATION

6. Testing and Evaluation

This chapter may include the following sections. (Students are required to perform the testing both manually and automatedly).

6.1 Manual Testing

Manual testing is a critical step in the evaluation process, as it validates system behaviour from the perspective of real end-users across different roles (Student, University, and Alumni). This method involves manually executing typical tasks, observing the system's output, and confirming that each function behaves as expected. Manual testing allows testers to replicate real-life usage patterns and uncover usability issues that automated tools might overlook. The system underwent manual testing to ensure smooth navigation, accurate data handling, and a seamless user experience for all user groups, including the integration of Lumina.

6.1.1: Manual Testing of Student as a user:

The Student Application underwent extensive manual testing to ensure that all core functionalities perform accurately and consistently. Each feature was tested individually by executing real-time actions and comparing the system's responses with the expected outcomes. The primary focus was to validate the user experience during authentication, profile management, uploading posts and interaction with other users.

Table 6.1: Manual testing of students as users

No.	Test case/Test script	Attribute value	Expected result	Actual Result	Result
1.	Verify user login after click on the 'Login' button on login form with correct input data	Username: StudentID101 Password: pass123	Successfully log into the main page of the system as Student.	Logged in successfully and role is student.	Pass
2.	Verify successful profile editing	Name: "Amir Khan"	Name updated successfully in the student	Name updated successfully in	Pass

	(Name change)		profile.	the user profile.	

6.1.2: Manual Testing of University as a user:

Manual testing for the University (Admin/FYP Committee) role focused on verifying each core feature used in daily administrative and management operations. The purpose of this testing was to ensure stable performance, accurate data handling, and smooth interaction with student and alumni data.

Table 6.2: Manual testing of university as users

No.	Test case/Test script	Attribute value	Expected result	Actual Result	Result
1.	Verify user login after click on the 'Login' button on login form with correct input data	Username: L001 Password: 1234	Successfully log into the main page of the system as University (FYP Committee) member.	Logged in successfully and role is university.	Pass
3.	Verify data listing display (e.g., students or alumni, all users list)	N/A	Accurate and complete list of student records/projects is displayed.	Accurate data display confirmed.	Pass

6.1.3:Manual Testing of Alumni as a user:

Manual testing for the Alumni application ensured reliable access to post-graduation resources, networking features, and specific Lumina channels relevant to career and networking opportunities. This testing phase confirmed that alumni can securely log in, update their career profiles, and access relevant institutional information without error.

Table 6.3: Manual testing of alumni as users

No.	Test case/Test script	Attribute value	Expected result	Actual Result	Result
1.	Verify user login after click on the 'Login' button on login form with correct input data	Username: AlumniID005 Password: AlumniPass	Successfully log into the main page of the system as Alumni.	Successfully log into the main page of the system as Alumni.	Pass

6.2 System testing

Once the system has been successfully developed, testing has to be performed to ensure that the system working as intended. This is also to check that the system meets the requirements stated earlier. Besides that, system testing will help in finding the errors that may be hidden from the user. There are few types of testing which includes the unit testing, functional testing and integration testing. The testing must be completed before it is being deploy for user to use.

6.3 Unit Testing

Unit Testing 1: Login as FYP Committee

Table 6.4: Login as FYP Committee Testing documents a successful test case (No. 1) confirming that the FYP Committee member can log into the main system page using the valid credentials Username: L001 and Password: 1234.

Testing Objective: To ensure the login form is working correctly

Table 6.4: Login as FYP Committee Testing

No.	Test case/Test script	Attribute value	Expected result	Actual Result	Result
1.	Verify user login after click on the „Login" button on login form with correct input data	Username: L001 Password: 1234	Successfully log into the main page of the system as FYP Committee member.	Logged in successfully and role is university.	Pass

Unit Testing 2: Edit Profile

Table 6.5: Edit Profile Testing documents two successful test cases for the Edit Profile feature, confirming that a user can update their Name (to "Ramisha Rahat") and Contact Number (to "03001234567") successfully.

Testing Objective: To ensure the edit profile form is working properly.

Table 6.5: Edit Profile Testing

No.	Test case/Test script	Attribute value	Expected result	Actual Result	Result
1.	Edit Name	Name: "Ramisha Rahat"	Name updated successfully in user profile	Name updated successfully in user profile	Pass
2.	Edit Contact Number	Phone: "03001234567"	Phone number updated successfully	Phone number updated successfully	Pass

6.4 Functional Testing

The functional testing will take place after the unit testing. In this functional testing, the functionality of each of the module is tested. This is to ensure that the system produced meets the specifications and requirements.

a-Functional Testing 1: Login with different roles

Table 6.5: Login with Different Roles Testing documents three successful Login with Different Roles test cases, confirming that FYP Committee, Student, and University Admin users are correctly authenticated and redirected to their respective main pages with the appropriate navigation bars.

Objective: To ensure that the correct page with the correct navigation bar is loaded.

Table 6.5: Login with Different Roles Testing

No.	Testcase/Test script	Attribute and value	Expected result	Actual Result	Result

1.	Login as a „FYP Committee" member.	Username: L001 Password: 1234	Main page for the FYP Committee member is loaded with the FYP Committee navigation bar	Successfully landed on main page as university but verification yet to pending.	Pass
2.	Login as a "Student"	Username: nihal121@gmail.com Password: 11111111	Main page for the student loaded with Student navigation bar	Successfully landed on main page as student.	Pass
3.	Login as a "University Admin"	Username: comsats@edu.pk Password: 11111111	Main page for Admin loaded with Admin navigation bar	Successfully landed on main page as university.	Pass

6.4-Integration Testing

Table 6.6: Integration Testing documents three successful Integration Test Cases, confirming that the FYP Committee member can log in, successfully upload student records for a project, and view their supervising student list.

Table 6.6: Integration Testing

No.	Testcase/Test script	Attribute and value	Expected result	Actual Result	Result
1.	Login as "FYP Committee" member	Username: L001 Password: 1234	Login successful and the FYP Committee page with its navigation bar is loaded and in the view profile page	Login successful and the FYP Committee page with its navigation bar is loaded and in the view profile page	Pass

2.	Upload student record for Project 1	-	File successfully uploaded and return to the upload page. Student records are updated.	File successfully uploaded and return to the upload page. Student records are updated.	Pass

6.5: Tools used:

Table 6.1 Tools Used documents the successful usage and integration of four development tools—Flutter, Firebase, Jupyter, and Web Scraping + Excel—across various test cases for UI, backend, data handling, and ML predictions, with all tests passing.

Table 6.1 Tools Used

Tool Name	**Tool Description**	**Applied on [list of related test cases / FR / NFR]**	**Results**
Flutter	Mobile app development framework	UI testing, Edit Profile, Login, Navigation, Profile Data display	Pass
Firebase	Backend services (Auth, Firestore, Storage, Push notifications)	Login, Profile updates, Upload student records, Notifications	Pass
Jupyter	Python notebook for data analysis and prototyping	Data processing, backend logic testing, ML-based predictions	Pass
Web Scraping + Excel	Collecting and storing structured data for testing	Importing student/project data for testing and validation	Pass

Chapter 7

CONCLUSION & FUTURE WORK

7. Conclusion and Future Work

7.1 Conclusion

The Lumina Career Counselling App serves as a modern solution to the long-standing challenges faced by students in navigating academic decisions, university admissions, and career planning. In a landscape where information is scattered, deadlines vary, and reliable guidance is often inaccessible, Lumina provides a unified, intelligent, and supportive platform. By integrating AI-driven recommendations with real-time updates, alumni mentorship, and mental-wellbeing support, the system ensures that students have all the tools they need to make confident, well-informed decisions.

The platform's architecture prioritizes both usability and security, creating a seamless experience for every user—whether student, alumnus, or university representative. Its design ensures scalability, efficient data management, and adherence to privacy standards, making the platform stable and dependable. By bridging communication gaps and offering personalized insights, Lumina significantly improves the traditional career counselling ecosystem, which is often manual, outdated, and inconsistent.

Overall, Lumina stands as a transformative step toward digitalizing career guidance in Pakistan. It not only centralizes essential academic information but also empowers learners through AI-powered mentorship and predictive analytics. As students continue to seek clarity and direction in an increasingly competitive world, Lumina ensures they are supported at every step of their academic and professional journey.

7.2 Future Work

Future enhancements of Lumina aim to further strengthen its impact and usability. These include:

- Expanding university coverage across all regions of Pakistan, providing students with more options and updated admission data.
- Improving AI accuracy for merit prediction and personalized recommendations, ensuring more reliable guidance.
- Scaling the app to support a larger user base, including advanced backend optimizations for performance.

- Adding AI-driven career suggestions and detailed skill mapping based on student profiles.
- Introducing video counselling sessions with alumni and career experts for interactive guidance.
- Integration with job placement portals and internship opportunities to provide a complete academic-to-career pathway.
- Publishing the app on official app stores with continuous updates and enhanced user experience.

These future developments will ensure Lumina evolves into a comprehensive, intelligent, and fully scalable career counselling platform, empowering students to make confident academic and professional career.

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